

DIGITALAIKART ORIGINAL

THE EXCEL JOB-READY PLAYBOOK

India Edition

From zero to passing the Excel test:
the 30-day system for analysts, MIS executives and every
job seeker who wrote 'Advanced Excel' on their resume

30

DAY PLAN

25+

FORMULAS TAUGHT

50

INTERVIEW Q&As

10

TIMED DRILLS

Every function and feature verified against Microsoft's official documentation

Built for the Indian job market · 2026-27 edition

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BEFORE YOU BEGIN

Introduction

Read This First

You wrote “Advanced Excel” on your resume. The job description said “proficiency in MS Excel”. Somewhere between those two lines sits a live test on a borrowed laptop, a messy sales file with 4,000 rows, and an interviewer with a stopwatch. This book is how you pass it.

What this playbook does

This is a job-readiness book, not a software manual. Microsoft's own documentation explains every button in Excel; free videos show you every menu. What is missing, and what you are paying for, is a **sequenced path**: what to learn first, what Indian interviewers actually test, how the live test feels, what a “good enough” answer sounds like, and how to prove the skill on a resume without a certificate.

Work through it with a spreadsheet open. Every example in this book is built around one small fictional company, Sharma Traders of Indore, so every formula you see is a formula you can type and verify in your own file within a minute.

Who it is for

- **Final-year students and freshers** applying for analyst, MIS, operations, finance and back-office roles.
- **Job seekers in a switch** — teachers, BPO associates, accountants, salespeople — moving into desk roles that test Excel.
- **Working professionals** who got hired on “basic Excel” and now sit in meetings where everyone else seems faster.
- **Anyone who has failed an Excel round** and wants to know exactly what went wrong.

How to use it

- 1 Read Chapters 1–2 today. They set up your practice file and your schedule.
- 2 Follow the 30-day plan in Chapter 3. Do not skip days; a missed day costs three on re-entry.
- 3 Chapters 4–14 are the skill itself: foundations, formulas, lookups, cleaning, pivots, charts, Power Query, shortcuts.
- 4 Chapters 15–19 are the conversion: interview questions, live-test tactics, AI tools, proof, and your first month at work.

The demand is real, and it is specific

Excel is not a glamorous skill, and that is exactly why it keeps working. A 2026 analysis of 93 job roles across eight Indian fields found Excel listed in 51 of them — the same league as SQL and data analysis, and ahead of most programming languages for non-IT roles. Indian job portals carry over three lakh live listings that mention Excel at any given time, from data entry to finance head. For MIS executive, analyst and back-office hiring, Excel is a screening round, not a bonus.

Hiring managers are consistent about what that screen checks. Recruiters for analyst roles describe Excel as non-negotiable at the first job: pivot tables, lookups and conditional logic, tested on a messy file, sometimes against a clock. Back-office and data-entry interviews routinely run four rounds — aptitude and typing, an MS Office/Excel test, HR, and an operations round — and the Excel round is where theory-only candidates fall

out. Interviewers themselves describe the trap questions they use: VLOOKUP versus INDEX-MATCH, broken absolute references, and building a SUMIFS for a described scenario rather than reciting its definition.

What this means for you

You are not competing with people who know Excel deeply. You are competing with people who memorised formula names. Practical, timed, hands-on preparation beats a certificate on LinkedIn almost every time — and this book is organised around exactly that.

Which roles, and what they pay

Reported fresher and early-career ranges from Indian salary trackers and training-industry surveys in 2026 (Indore-to-Bengaluru spread; actual offers vary widely by city and company):

Role	Excel they expect	Typical reported range
MIS Executive	Pivots, VLOOKUP, charts, basic macros	Rs 15,000–35,000 / month
Data Analyst (entry)	Pivots, lookups, Power Query + one BI tool	Rs 5–18 LPA
Finance / Accounts Exec.	IF logic, financial formulas, reports	Rs 15,000–30,000 / month
HR / Admin Executive	VLOOKUP, conditional formatting, payroll sheets	Rs 12,000–25,000 / month
Operations Analyst	Dashboards, Power Query	Rs 20,000–45,000 / month

The pattern worth noticing: Excel matters most at the **start** of a data career. It is the door. SQL, Power BI and Python decide how far you go after you are inside, but the entry test is Excel almost everywhere, because a large share of real analysis in Indian services firms, consulting and banking still happens in a spreadsheet.

What the test actually looks like

Excel assessments in Indian hiring fall into three formats, and you will usually face the second:

Format	What happens	Where it is used
Multiple choice	Definitions and shortcut questions, 15–30 items, online	Screening for BPO and admin roles
Practical live test	A real spreadsheet; you clean data, write lookups, build a pivot, make a chart — often timed	Analyst, MIS, operations, finance
Take-home task	Raw data given; you return a one-page analysis or dashboard	Senior analyst and product roles

The practical test almost always uses the same skeleton: a raw export with dirty data — extra spaces, mixed date formats, duplicates, text-stored-as-numbers — followed by five to eight tasks of increasing difficulty: clean it, summarise it with a pivot, add a lookup from a second sheet, a conditional count, one chart, and a short sentence on what the numbers say. If you can do that sequence in under 30 minutes, you will pass most live rounds. That sequence is what this book trains.

Skill by job level

Target role	Level you need	Core skills tested
Data entry, admin	Basic	SUM, COUNT, sort, filter, formatting
Sales exec, HR, ops	Intermediate	IF, VLOOKUP, SUMIF, basic pivots, charts
MIS exec, finance analyst	Intermediate-plus	INDEX-MATCH, SUMIFS, COUNTIFS, pivot charts, dashboards
Data analyst, senior finance	Advanced	Power Query, macros, advanced pivots, complex models

Be honest about which row you are applying for, and prepare one level above it. Interviewers rarely fail candidates for being slow; they fail them for not knowing the function their daily work depends on.

What you need (and what you do not)

You need a working copy of Excel and 45 minutes a day. You do not need a paid course, a new laptop, or any prior experience.

Which Excel version to use

Four versions matter in India, and they differ in ways an interviewer can see:

Version	Where you meet it	What is missing
Excel 2016 / 2019	Most offices and exam-centre machines	No XLOOKUP, no dynamic arrays
Excel 2021 (perpetual)	Smaller firms, one-time licence	Has XLOOKUP and dynamic arrays; no Copilot
Microsoft 365 (subscription)	Mid and large companies, the default	Everything: XLOOKUP, Copilot, newest functions
Excel for the web (free)	Any browser, a free Microsoft account	Core functions, pivots and charts; some limits

Free access

Excel for the web is free at office.com with a Microsoft account, and it runs the formulas, pivot tables and charts this book teaches. If you cannot install Excel, practice there. If you use Google Sheets at home, everything in Chapters 4–9 translates directly (VLOOKUP, SUMIFS, INDEX-MATCH all exist); Sheets calls Power Query “power tools” differently and Copilot does not apply.

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XLOOKUP works in Microsoft 365, Excel 2021 and Excel 2024; it does not exist in Excel 2016 or 2019. Before a live test, check which version the company runs — ask HR, or check File > Account. If the machine runs Office 2016 and you write =XLOOKUP(...), you get a #NAME? error and lose the task. The safe answer in an interview: “I use XLOOKUP on Microsoft 365, but I can write VLOOKUP or INDEX-MATCH when the file has to work on older versions.” That one sentence signals more competence than any certificate.

Build the Sharma Traders practice file

Every exercise in this book uses one workbook you will build yourself in fifteen minutes. Creating it is your first exercise.

- 1 Open a new workbook. Rename Sheet1 to **Sales** and Sheet2 to **Targets**.
- 2 In Sales, enter these headers in row 1: Date, Region, Salesman, Product, Qty, Unit Price, Amount, Paid.
- 3 Enter 20 rows of sales data across October–December 2026: dates in column A (mix formats on purpose — some as 05-10-2026, some as 05/10/2026); regions North, South, East, West; four salesmen (Aarav, Diya, Kabir, Meera); three products (Desk, Chair, Lamp); Qty between 1 and 20; Unit Price between 1,200 and 9,500.
- 4 Amount column: leave it empty for now — you will fill it with your first formula in Chapter 6. Paid column: type Yes or No, with three rows left blank.
- 5 In Targets, create a two-column table: Region (North, South, East, West) and Target (4,00,000; 3,00,000; 2,50,000; 3,50,000).
- 6 Deliberately enter two duplicate rows in Sales (same date, salesman, product, qty) and type one salesman name as “kabir ” with a trailing space. Name one product “chair” in lowercase.

Yes, the file is intentionally dirty. Chapters 8 and 9 exist because of it.

The 30-day schedule

Forty-five minutes daily beats four hours on Sunday — Excel is a motor skill as much as a knowledge skill. Each day below maps to chapters of this book; when a day says “drill”, do it against the stopwatch in Chapter 16.

Days	Focus	Chapters
1–2	Why Excel hires; set up; build the practice file	1–2
3–5	Cells, ranges, references, formatting, tables, print setup	4–5
6–10	The core twelve formulas; text, dates, error handling	6, 8
11–13	Lookups: VLOOKUP, XLOOKUP, INDEX-MATCH; lookup drills	7
14–16	Data cleaning; the full clean of Sharma Traders	9
17–20	Pivot tables and pivot charts, every angle	10
21–23	Charts and the one-page dashboard build	11
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28	Shortcuts under speed pressure	14
29	Interview questions, first 25, out loud	15
30	Timed mock: full live-test simulation, then review	16

If you have seven days, not thirty

The exam-school version: Day 1 foundations and formatting (Ch 4–5). Day 2 core formulas (Ch 6). Day 3 lookups (Ch 7). Day 4 cleaning (Ch 9). Day 5 pivots (Ch 10). Day 6 charts (Ch 11) and shortcuts (Ch 14). Day 7 one full timed mock plus the first 20 interview questions (Ch 15–16). It is survivable. The 30-day version is simply better.

The grid, and the words interviewers use

A **workbook** is the file. A **worksheet** (or sheet) is one tab inside it. The **active cell** is the outlined one; its name — C7, meaning column C, row 7 — sits in the Name Box to the left of the formula bar. A **range** is a rectangle of cells written with a colon: A1:C10 means everything from A1 to C10. That is the whole vocabulary; interviewers ask “what is the difference between a workbook and a worksheet” in fresher rounds precisely to check you know even this much.

What lives in a cell

Excel stores numbers, text, dates, times, TRUE/FALSE values, and formulas. Two traps follow from this. First, **dates are numbers** — 01-01-1900 is 1, and 30-09-2026 is roughly 46,266 — which is why you can subtract two dates to get days, and why a date that shows left-aligned like text is broken. Second, **numbers can be stored as text** when exported from ERP or pasted from PDF; SUM ignores them, and the tell is a small green triangle plus a right-alignment that has become left. Interviewers hand you files with exactly this defect.

Entering and editing

- Press Enter to confirm and move down; Tab to confirm and move right; Escape to cancel.
- Double-click a cell (or press F2) to edit inside it; press F2 after a formula to see colour-coded references.
- Ctrl+Z undo, Ctrl+Y redo — the two most-used keys in any live test.

- Start a cell with an apostrophe (') to force text — typing 001 as a number drops the zeros; '001 keeps them.
- Ctrl+Enter fills the selection with what you typed without moving the cursor; Ctrl+D copies the cell above down.

References: the single most tested concept

When a formula says =B2+C2 and you copy it down one row, Excel does not copy it literally — it rewrites it to =B3+C3. That is a **relative reference**: the formula remembers positions, not addresses. Press F4 while editing a reference to cycle through the four forms:

Written as	Means	When copied right	When copied down
B2 (relative)	Two cells to the left of me, one up	Shifts to C2	Shifts to B3
\$B2 (column locked)	Column B is fixed; row floats	Stays B2	Shifts to B3
B\$2 (row locked)	Row 2 is fixed; column floats	Shifts to C2	Stays B2
\$B\$2 (absolute)	That exact cell, always	Stays B2	Stays B2

The classic interview trap is a broken lookup: =VLOOKUP(A2,F2:G20,2,FALSE) dragged down works fine because both references grow together — but =A2*\$E\$1 converted to a tax-rate formula without the dollars will silently point at empty cells and return zeros. When an interviewer slides you a sheet full of #REF! or zero errors and asks “why is this broken”, the answer is almost always “the dollar signs are missing or wrong”. Say that, then fix it live.

The tax-rate rule

If a constant lives in one cell that many formulas need — a tax rate, a conversion factor, a target — lock it with \$ on both sides: \$E\$1. Type the dollars while writing the formula; do not come back later.

Select like a professional

- Click a cell; Ctrl+A inside data selects the whole current region; Ctrl+A again selects the sheet.
- Ctrl+Arrow jumps to the edge of the data — Ctrl+Down from the first data row lands on the last non-empty one. The fastest way to see how many rows you actually got.
- Ctrl+Shift+Arrow extends the selection to that edge. Select-and-drag over 4,000 rows marks you as a beginner in a live test.
- Click one cell in a header row, then Ctrl+Shift+L toggles filters; Ctrl+Shift+L again removes them.
- Name Box trick: type A2:A5000 in the Name Box and press Enter to select an exact range by typing.

Sheets: the habits that make files survivable

- Double-click a tab to rename it; a tab called “Sheet1” in a shared file is how people lose work.
- Right-click a tab for Move or Copy (to duplicate a sheet — the honest way to keep a backup before a big edit), tab colours, and Hide.
- Ctrl+Page Down / Ctrl+Page Up moves between sheets without the mouse.
- Never leave a “check” of merged cells inside a data table. Merge and Center is for a report title, never for data you will sort, filter, pivot or look up — merged cells break all four.

The interview one-liner

“Raw data lives in its own sheet exactly as it arrived; every calculation and pivot sits in separate sheets that read from it — so refreshing the raw data refreshes everything.” This sentence, delivered while opening your practice file, is what ‘good Excel hygiene’ sounds like to a hiring manager.

Formatting is credibility

Nobody fails an Excel test because their chart was two shades off-brand. People fail credibility checks: numbers that do not add up because a column was never widened, dates that read #####, decimals that disagree between rows, totals that were typed by hand instead of calculated. Formatting, done the way this chapter teaches, is how a reviewer decides in ten seconds whether to trust the file.

Number formats that matter

- Ctrl+1 opens Format Cells for anything selected — the single most useful formatting keystroke.
- Indian format for currency: Format Cells > Number > Currency, symbol Rs., and tick the lakhs/crores grouping under the Indian number format option (Excel calls it the Indian grouping in regional settings; if unavailable, a custom format does it — see below).
- Custom Indian comma format: Format Cells > Custom and enter `[>=10000000]##\,##\,##\,##0;[>=100000]##\,##\,##0;##,##0` — this renders 12345678 as 1,23,45,678.
- Percent format multiplies by 100 for display: typing 12% stores 0.12. A growth column where 30% and 0.3 coexist has mixed formats — fix it before anyone notices, because someone always notices.
- Dates: pick one display format for the whole file (DD-MM-YYYY is the Indian convention) and never mix 05-10-26 with 05/10/26 in the same column.

The look of a clean table

- Bold headers, one background fill (light grey or pale blue), thin borders — or no borders at all with banded rows.
- Numbers right-aligned (default), text left-aligned (default). If a number sits left-aligned, it is text — investigate.
- One decimal convention across the file: two decimals for money, none for counts.
- Column widths wide enough that nothing shows #####; wrap text (Alt+Enter for a manual line break inside a cell) instead of shrinking fonts.

Format as Table: the feature people under-use

Select your data and press Ctrl+T. Excel converts the range into a **Table** (not to be confused with a pivot table — different feature, same unfortunate word). What you get: banded rows, filter arrows, a name (Table1) you can reference in formulas, formulas that auto-fill down the entire column when you type one, and ranges that grow automatically when new rows are appended. In Chapter 10, pivot tables built on a Table pick up new data on refresh without you ever re-selecting the range.

Formulas written inside a Table switch to structured references — Sales[Qty] instead of \$E\$2:\$E\$1000 — which read like sentences and scare people who have not seen them. They behave exactly like ordinary references. The interview-safe posture: know Ctrl+T, use it for your own files, recognise structured references when you inherit them.

Interview question you will get

“What is the difference between a range and a Table?” Answer: a Table is a named, self-expanding range with built-in filtering and styling; formulas auto-fill, references stay correct as rows are added, and charts or pivots built on it grow with the data. A plain range does none of that.

Freeze, split, and the view menu

- View > Freeze Panes > Freeze Top Row keeps headers visible while you scroll 4,000 rows — do this in the first minute of any live test on a long file.
- Freeze First Column for wide files; Freeze Panes with a custom cell selected freezes everything above and left of that cell.
- Window > Split divides the window into scrollable panes when comparing top and bottom of the same sheet.
- Hide a row or column with Ctrl+9 (rows) and Ctrl+0 (columns) — and remember to unhide before submitting a test file; hidden rows in a submitted file look like fudged data.

Print setup: the polite finish

Analyst tests sometimes end with “set this up to print” or “save as PDF”. The 60-second version:

- 1 Page Layout > Print Titles: set Rows to repeat at top = row 1, so every printed page carries the headers.
- 2 Fit Sheet on One Page (or fit to 1 page wide) from the Scale to Fit group — one dimension only; two shrinks the font to ant size.
- 3 Set orientation: Wide for files with more than 8 columns.
- 4 Margins > Narrow, and Center on Page: Horizontally.
- 5 Ctrl+P to preview, then Print to PDF rather than paper.

Shortcut or path	What it does
Ctrl+1	Format Cells dialog
Ctrl+T	Convert range to Table
Ctrl+9 / Ctrl+0	Hide selected rows / columns
Shift+Ctrl+9 / 0	Unhide rows / columns
Alt+Enter	Line break inside a cell
Ctrl+Shift+1 / 2 / 3 / 4 / 5	Number / Time / Date / Currency / Percent formats
Page Layout tab	Print titles, scale, margins

THE MARKET

Chapter 1

Why Excel Still Decides Who Gets Hired

The demand is real, and it is specific

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Take-home task	Raw data given; you return a one-page analysis or dashboard	Senior analyst and product roles

The practical test almost always uses the same skeleton: a raw export with dirty data — extra spaces, mixed date formats, duplicates, text-stored-as-numbers — followed by five to eight tasks of increasing difficulty: clean it, summarise it with a pivot, add a lookup from a second sheet, a conditional count, one chart, and a short sentence on what the numbers say. If you can do that sequence in under 30 minutes, you will pass most live rounds. That sequence is what this book trains.

Skill by job level

Target role	Level you need	Core skills tested
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Be honest about which row you are applying for, and prepare one level above it. Interviewers rarely fail candidates for being slow; they fail them for not knowing the function their daily work depends on.

YOUR TOOLKIT

Chapter 2

Get Set Up in One Evening

What you need (and what you do not)

You need a working copy of Excel and 45 minutes a day. You do not need a paid course, a new laptop, or any prior experience.

Which Excel version to use

Four versions matter in India, and they differ in ways an interviewer can see:

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Yes, the file is intentionally dirty. Chapters 8 and 9 exist because of it.

THE SCHEDULE

Chapter 3

The 30-Day Job-Ready Plan

The 30-day schedule

Forty-five minutes daily beats four hours on Sunday — Excel is a motor skill as much as a knowledge skill. Each day below maps to chapters of this book; when a day says “drill”, do it against the stopwatch in Chapter 16.

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If you have seven days, not thirty

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FOUNDATIONS

Chapter 4

Cells, Ranges and References

The grid, and the words interviewers use

A **workbook** is the file. A **worksheet** (or sheet) is one tab inside it. The **active cell** is the outlined one; its name — C7, meaning column C, row 7 — sits in the Name Box to the left of the formula bar. A **range** is a rectangle of cells written with a colon: A1:C10 means everything from A1 to C10. That is the whole vocabulary; interviewers ask “what is the difference between a workbook and a worksheet” in fresher rounds precisely to check you know even this much.

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Written as	Means	When copied right	When copied down
B2 (relative)	Two cells to the left of me, one up	Shifts to C2	Shifts to B3
\$B2 (column locked)	Column B is fixed; row floats	Stays B2	Shifts to B3
B\$2 (row locked)	Row 2 is fixed; column floats	Shifts to C2	Stays B2
\$B\$2 (absolute)	That exact cell, always	Stays B2	Stays B2

The classic interview trap is a broken lookup: =VLOOKUP(A2,F2:G20,2,FALSE) dragged down works fine because both references grow together — but =A2*\$E\$1 converted to a tax-rate formula without the dollars will silently point at empty cells and return zeros. When an interviewer slides you a sheet full of #REF! or zero errors and asks “why is this broken”, the answer is almost always “the dollar signs are missing or wrong”. Say that, then fix it live.

The tax-rate rule

If a constant lives in one cell that many formulas need — a tax rate, a conversion factor, a target — lock it with \$ on both sides: \$E\$1. Type the dollars while writing the formula; do not come back later.

Select like a professional

- Click a cell; Ctrl+A inside data selects the whole current region; Ctrl+A again selects the sheet.
- Ctrl+Arrow jumps to the edge of the data — Ctrl+Down from the first data row lands on the last non-empty one. The fastest way to see how many rows you actually got.
- Ctrl+Shift+Arrow extends the selection to that edge. Select-and-drag over 4,000 rows marks you as a beginner in a live test.
- Click one cell in a header row, then Ctrl+Shift+L toggles filters; Ctrl+Shift+L again removes them.
- Name Box trick: type A2:A5000 in the Name Box and press Enter to select an exact range by typing.

Sheets: the habits that make files survivable

- Double-click a tab to rename it; a tab called “Sheet1” in a shared file is how people lose work.
- Right-click a tab for Move or Copy (to duplicate a sheet — the honest way to keep a backup before a big edit), tab colours, and Hide.
- Ctrl+Page Down / Ctrl+Page Up moves between sheets without the mouse.
- Never leave a “check” of merged cells inside a data table. Merge and Center is for a report title, never for data you will sort, filter, pivot or look up — merged cells break all four.

The interview one-liner

“Raw data lives in its own sheet exactly as it arrived; every calculation and pivot sits in separate sheets that read from it — so refreshing the raw data refreshes everything.” This sentence, delivered while opening your practice file, is what ‘good Excel hygiene’ sounds like to a hiring manager.

FOUNDATIONS

Chapter 5

Formatting That Earns Trust

Formatting is credibility

Nobody fails an Excel test because their chart was two shades off-brand. People fail credibility checks: numbers that do not add up because a column was never widened, dates that read #####, decimals that disagree between rows, totals that were typed by hand instead of calculated. Formatting, done the way this chapter teaches, is how a reviewer decides in ten seconds whether to trust the file.

Number formats that matter

- Ctrl+1 opens Format Cells for anything selected — the single most useful formatting keystroke.
- Indian format for currency: Format Cells > Number > Currency, symbol Rs., and tick the lakhs/crores grouping under the Indian number format option (Excel calls it the Indian grouping in regional settings; if unavailable, a custom format does it — see below).
- Custom Indian comma format: Format Cells > Custom and enter `[>=10000000]##\,##\,##\,##0;[>=100000]##\,##\,##0;##,##0` — this renders 12345678 as 1,23,45,678.
- Percent format multiplies by 100 for display: typing 12% stores 0.12. A growth column where 30% and 0.3 coexist has mixed formats — fix it before anyone notices, because someone always notices.
- Dates: pick one display format for the whole file (DD-MM-YYYY is the Indian convention) and never mix 05-10-26 with 05/10/26 in the same column.

The look of a clean table

- Bold headers, one background fill (light grey or pale blue), thin borders — or no borders at all with banded rows.
- Numbers right-aligned (default), text left-aligned (default). If a number sits left-aligned, it is text — investigate.
- One decimal convention across the file: two decimals for money, none for counts.
- Column widths wide enough that nothing shows #####; wrap text (Alt+Enter for a manual line break inside a cell) instead of shrinking fonts.

Format as Table: the feature people under-use

Select your data and press Ctrl+T. Excel converts the range into a **Table** (not to be confused with a pivot table — different feature, same unfortunate word). What you get: banded rows, filter arrows, a name (Table1) you can reference in formulas, formulas that auto-fill down the entire column when you type one, and ranges that grow automatically when new rows are appended. In Chapter 10, pivot tables built on a Table pick up new data on refresh without you ever re-selecting the range.

Formulas written inside a Table switch to structured references — Sales[Qty] instead of \$E\$2:\$E\$1000 — which read like sentences and scare people who have not seen them. They behave exactly like ordinary references. The interview-safe posture: know Ctrl+T, use it for your own files, recognise structured references when you inherit them.

Interview question you will get

“What is the difference between a range and a Table?” Answer: a Table is a named, self-expanding range with built-in filtering and styling; formulas auto-fill, references stay correct as rows are added, and charts or pivots built on it grow with the data. A plain range does none of that.

Freeze, split, and the view menu

- View > Freeze Panes > Freeze Top Row keeps headers visible while you scroll 4,000 rows — do this in the first minute of any live test on a long file.
- Freeze First Column for wide files; Freeze Panes with a custom cell selected freezes everything above and left of that cell.
- Window > Split divides the window into scrollable panes when comparing top and bottom of the same sheet.
- Hide a row or column with Ctrl+9 (rows) and Ctrl+0 (columns) — and remember to unhide before submitting a test file; hidden rows in a submitted file look like fudged data.

Print setup: the polite finish

Analyst tests sometimes end with “set this up to print” or “save as PDF”. The 60-second version:

- 1 Page Layout > Print Titles: set Rows to repeat at top = row 1, so every printed page carries the headers.
- 2 Fit Sheet on One Page (or fit to 1 page wide) from the Scale to Fit group — one dimension only; two shrinks the font to ant size.
- 3 Set orientation: Wide for files with more than 8 columns.
- 4 Margins > Narrow, and Center on Page: Horizontally.
- 5 Ctrl+P to preview, then Print to PDF rather than paper.

Shortcut or path	What it does
Ctrl+1	Format Cells dialog
Ctrl+T	Convert range to Table
Ctrl+9 / Ctrl+0	Hide selected rows / columns
Shift+Ctrl+9 / 0	Unhide rows / columns
Alt+Enter	Line break inside a cell
Ctrl+Shift+1 / 2 / 3 / 4 / 5	Number / Time / Date / Currency / Percent formats
Page Layout tab	Print titles, scale, margins

FORMULAS

Chapter 6

The Core Formulas Every Test Asks For

How formulas think

Every formula starts with `=`. What follows is an expression Excel evaluates to a single answer, using cells' **values** (what the cell displays) and cells' **positions** (where the formula found them). When you copy a formula, positions shift (Chapter 4); values follow the cell. Master that one sentence and half of Excel's confusing behaviour — why copying breaks some formulas, why a lookup returns a stale answer after sorting — becomes predictable.

Function names are not case-sensitive (`=sumifs` works), but write them in capitals anyway; reviewers read formulas, and reviewers are human. Arguments are separated by commas in the Indian/US setting (your Windows region decides this; if your Excel expects semicolons, the Windows list separator is set to a semicolon — change it in Windows Region settings or just go with it).

The arithmetic core

SUM

`=SUM(D2:D21)` adds the range

SUM adds numbers and skips text. Press `Alt+=` (Alt and Equals together) with a cell below your data and Excel writes `=SUM()` around the numbers it guesses you mean — it is right most of the time. AVERAGE, MAX and MIN work identically and answer the questions every dashboard starts with.

AVERAGE

`=AVERAGE(D2:D21)` arithmetic mean

MAX

`=MAX(D2:D21)` largest value

MIN

`=MIN(D2:D21)` smallest value

COUNT family — know all four or lose easy marks

Function	Counts	In Sharma Sales: counts the...
<code>=COUNT(D2:D21)</code>	Cells with numbers	Qty cells (numeric)
<code>=COUNTA(A2:A21)</code>	Cells with any content	All filled rows incl. text
<code>=COUNTBLANK(H2:H21)</code>	Empty cells	Rows where Paid is blank
<code>=COUNTIF(E2:E21,"chair")</code>	Cells meeting one condition	Rows where product = chair

Interview favourite

The difference between COUNT and COUNTA comes up in almost every fresher round. One line: COUNT counts numbers only; COUNTA counts anything non-empty. Then prove it live on two ranges.

IF and its family: conditional logic

IF

```
=IF(H2="Yes", "Paid", "Pending")
```

IF takes three parts: a condition, what to show when true, what to show when false. Conditions are built from comparisons (=, <> for not-equal, >, <, >=, <=) and can be combined with AND (all must hold) and OR (any must hold):

IF + AND

```
=IF(AND(G2>50000, H2="Yes"), "VIP", "Standard")
```

Nested IFs — an IF inside an IF — grade values into bands, and interviewers love them because they reveal whether you can hold a decision tree in your head:

Nested IF

```
=IF(G2>=100000, "A", IF(G2>=50000, "B", IF(G2>=10000, "C", "D")))
```

Read it aloud: if amount is at least one lakh, A; otherwise if at least fifty thousand, B; otherwise if at least ten thousand, C; else D. On Microsoft 365 you can write the same thing with IFS — =IFS(G2>=100000,"A", G2>=50000,"B", G2>=10000,"C", TRUE,"D") — flatter and easier to audit, but not available on Excel 2016, so know nested IF for the test and IFS for the job.

The banding trap

A common live-test task: “tag each order Small / Medium / Large”. The mistake is writing =IF(G2>=10000,"Medium",...) without covering the below-10000 case first, or overlapping bands. Write the bands on paper before the formula: below 10k, 10k to 50k, 50k to 1 lakh, above 1 lakh. Then each IF asks exactly one question.

SUMIF, SUMIFS and their cousins

SUMIF

```
=SUMIF(C2:C21, "Kabir", G2:G21)
```

SUMIF reads: look in the salesmen column for Kabir, and sum the matching amounts. The first argument is where to look, second is what to look for, third is what to add up. Note the third argument is optional if the range you search is the range you add.

SUMIFS

```
=SUMIFS(G2:G21, C2:C21, "Kabir", B2:B21, "North")
```

SUMIFS is SUMIF with the arguments reordered for multiple conditions: **first what to add, then pairs of range-and-condition**. This order swap is Microsoft's design (confirmed in their documentation), it is the single most common formula error in tests, and admitting you double-check it every time is a legitimate answer when an interviewer asks how you avoid mistakes. Up to 127 condition pairs are allowed — you will use two or three.

Task in the test	Formula on Sharma Sales
Total amount for chairs	=SUMIF(E2:E21,"chair",G2:G21)
Kabir's sales in the North	=SUMIFS(G2:G21,C2:C21,"Kabir",B2:B21,"North")
Number of orders above Rs 50,000	=COUNTIF(G2:G21,">50000")
Large UNPAID orders (count)	=COUNTIFS(G2:G21,">=50000",H2:H21,"No")
Average order in the South	=AVERAGEIF(B2:B21,"South",G2:G21)

Quotation marks around conditions

When the condition is a comparison or text, it goes in quotes: ">50000", "Kabir". =COUNTIF(G2:G21, >50000) without quotes is the error reviewers watch for. Numbers alone can go bare: 50000 means equal to 50000.

Rounding and SUBTOTAL

ROUND

=ROUND(G2*0.18, 2)

ROUND takes digits to keep: 2 for paise, 0 for whole rupees. ROUNDUP and ROUNDDOWN exist; so does MROUND for rounding to a multiple (pack sizes). Percentages that show 33.333333% on a report are a rounding discipline failure, not an Excel failure — fix display with ROUND, not by hoping the format hides it.

SUBTOTAL

=SUBTOTAL(109, G2:G21)

SUBTOTAL with 9 (or 109) sums only the **visible** rows — filter the data and the total follows the filter. In any test where you filter and are asked “what is the total of what we are looking at”, SUBTOTAL is the expected answer; SUM keeps counting the hidden rows and is the trap.

Absolute references in real formulas

The pattern that appears in a majority of live tests: a rate that sits in one cell and a column that must all use it. On Sharma Traders, put 18% GST in cell J1, then in a new column:

The dollar-sign pattern

=ROUND(G2*\$J\$1, 2)

Copy it down 20 rows and every row correctly reads the same J1. Remove the dollars and row 5 computes G5*J5 — empty — giving zeros. If you internalise one interaction between chapters, make it this one plus F4.

Five business questions, five formulas

Formulas stick when they answer a question rather than fill a syllabus. These five cover the shape of most test tasks; each is written for the Sharma Sales sheet (columns: A Date, B Region, C Salesman, D Product, E Qty, F Unit Price, G Amount, H Paid, GST rate in J1).

1. What did Kabir actually sell, net of the orders that are not paid yet?

SUMIFS

=SUMIFS(G2:G21, C2:C21, "Kabir", H2:H21, "Yes")

Two conditions, one formula. The sum range comes first; each condition follows as a range-criteria pair. If the test adds “and only desks”, a third pair does it — no restructuring needed.

2. Which orders are big enough to need the manager's sign-off?

Nested IF

```
=IF(G2>=50000, "Manager approval", IF(G2>=20000, "Lead approval", "Auto"))
```

Two thresholds, three outcomes. The discipline is the order: check the highest band first, because 75,000 is also >=20,000 and would otherwise stop at the lower band. Banding questions in tests are really order-of-tests questions.

3. What share of revenue came from the North region?

SUMIF / SUM

```
=SUMIF(B2:B21, "North", G2:G21) / SUM(G2:G21)
```

Part over whole, formatted as percent. The pivot version is Show Values As > % of Grand Total; the formula version is what you write when the answer must live in a sentence or a summary cell.

4. What is the average order value of UNPAID orders?

AVERAGEIFS

```
=AVERAGEIFS(G2:G21, H2:H21, "No")
```

AVERAGEIFS takes the same range-pair structure as SUMIFS. The trap: unpaid rows with a blank (not “No”) are excluded — after Chapter 9's fill-blanks move, they are the same thing. Consistency of the Paid column is part of the answer.

5. How many days old is each order, and which are overdue by 30+ days?

TODAY with IF

```
=TODAY()-A2 and =IF(TODAY()-A2>30, "Overdue", "OK")
```

Dates are day-numbers, so subtraction is elapsed days; TODAY() moves with the clock, so the column recomputes every time the file opens. This pair is the seed of every collections report you will ever build, and interviewers know it.

FORMULAS

Chapter 7

Lookups Done Right

Why lookups decide interviews

A lookup answers: here is a key (employee ID, product, region), give me the matching detail from another table. Three functions do this job — VLOOKUP, XLOOKUP and INDEX-MATCH — and interviewers use them to separate candidates more than any other family, because they map directly onto daily work: pull the target from the Targets sheet against the region in the Sales sheet.

VLOOKUP: the one everybody asks about first

VLOOKUP

```
=VLOOKUP(B2, Targets!$A$2:$B$5, 2, FALSE)
```

Read the four arguments: what to look up (the region in B2); the table to search (Targets sheet, columns A to B); which column of that table to return (2 = the second column, Target); and the match type — FALSE means exact match. Always FALSE. Leave it blank or write TRUE and Excel assumes an **approximate** match, which requires sorted data and silently returns a near-miss instead of an error. Microsoft's own documentation warns that approximate is the default when you omit the argument — the most dangerous default in Excel.

The two VLOOKUP limits interviewers test

- It can only look **right**: the lookup column must be the first column of the table range. Need a value to the left? VLOOKUP cannot; INDEX-MATCH or XLOOKUP can.
- Insert a column inside the table and the hardcoded 3 for column 3 now points at the wrong data — no error, just quietly wrong answers. This is why reviewers prefer XLOOKUP or INDEX-MATCH on inherited files.

The exact-match error

VLOOKUP + IFERROR

```
=IFERROR(VLOOKUP(B2, Targets!$A$2:$B$5, 2, FALSE), 0)
```

When VLOOKUP finds nothing it returns #N/A — for instance looking up a region typed as “north ” with a trailing space. IFERROR wraps the lookup and returns something civilized instead. Be careful with it: #N/A is a signal that the data has a problem; blindly wrapping every lookup in IFERROR and returning 0 is how wrong numbers reach a client's slide. The interview answer: “I wrap IFERROR for display, but I first check why the match failed — usually a trailing space or a number stored as text.”

XLOOKUP: the modern standard

XLOOKUP

```
=XLOOKUP(B2, Targets!$A$2:$A$5, Targets!$B$2:$B$5, "No target")
```

XLOOKUP takes: what to look up, where to look, what to return, and optionally what to show if not found. Compare that with VLOOKUP: no column counting, the return column can be to the **left** of the search column, no table-array to break when columns are inserted, and if-not-found is a built-in argument instead of a wrapping IFERROR. Microsoft's documentation is blunt about availability: XLOOKUP exists in Microsoft 365, Excel 2021 and Excel 2024, and does **not** exist in Excel 2016 or 2019.

Two more optional arguments matter: match mode (0 exact is the default; -1 exact-or-next-smaller; 1 exact-or-next-larger; 2 wildcard) and search mode (-1 searches from the end — the elegant way to find the **latest** transaction for a customer, a favourite MIS interview twist).

The version question, answered for the test

Say this in the interview

"XLOOKUP is my default on Microsoft 365. On files that must run on Excel 2016 or 2019 I fall back to VLOOKUP with IFERROR for exact matches, or INDEX-MATCH when the return column sits to the left." This answer covers versions, limits and fallbacks in two sentences and no interviewer pushes further.

INDEX and MATCH: the classic pair

INDEX + MATCH

```
=INDEX(Targets!$B$2:$B$5, MATCH(B2, Targets!$A$2:$A$5, 0))
```

Read it inside-out. MATCH finds where the region sits in the list of regions — position 3, say. INDEX returns the value at position 3 of the results column. Together: any column, any direction, no counting, immune to inserted columns, works in every Excel version since the 1990s. It is the answer when an interviewer asks "how would you look up a value to the left?" — the single most-quoted interview question in the Excel canon.

Two-way lookup: the senior move

Two-way lookup

```
=INDEX(Sales!$F$5:$M$16, MATCH("Kabir", Sales!$E$5:$E$16, 0), MATCH("Nov", Sales!$F$4:$M$4, 0))
```

INDEX with two MATCHes — one for the row, one for the column — returns the value where a row label crosses a column label, e.g. Kabir's row against the November column of a monthly matrix. Mentioning that you can do this, then drawing the 2x2 on paper, is worth more in an interview than any certificate.

Choosing between them

Situation	Use	Why
Company runs Microsoft 365 / 2021+	XLOOKUP	Cleanest syntax, left-lookups, built-in not-found
Old Excel 2016/2019 machine	VLOOKUP (FALSE) or INDEX-MATCH	XLOOKUP returns #NAME?
Return column is left of search column	INDEX-MATCH or XLOOKUP	VLOOKUP cannot look left
Inheriting a fragile file, may change	INDEX-MATCH	Survives inserted columns
Need the latest record per customer	XLOOKUP search -1	Reverse search built in

Why #N/A happens — the five causes

1) trailing or double spaces — fix with TRIM; 2) number stored as text on one side — fix with VALUE or multiply-by-1; 3) region set to display Indian date but stored differently; 4) the value truly does not exist — business answer, not Excel answer; 5) approximate-match VLOOKUP on unsorted data — switch to FALSE. Diagnosing #N/A out loud, calmly, is the strongest thing you can do in a live test.

FORMULAS

Chapter 8

Text, Dates and Error-Reading

Text functions: what cleaning really uses

Between 60 and 80 percent of an analyst's first-year Excel time goes into re-shaping text and dates that arrived ugly. These are the functions tests and real work actually call on.

Task	Formula	On what input
First name from full name	=LEFT(A2, FIND(" ", A2)-1)	"Aarav Sharma" → "Aarav"
Last name	=RIGHT(A2, LEN(A2)-FIND(" ", A2))	"Aarav Sharma" → "Sharma"
Middle slice	=MID(A2, 6, 4)	Characters 6-9 of a code
Kill stray spaces	=TRIM(A2)	"kabir " → "kabir"
Length check	=LEN(A2)	Spots double spaces: length is off
Join cells	=CONCAT(B2, " - ", C2) or =TEXTJOIN(" ", TRUE, B2:C2)	First + last name
Recode case	=PROPER(A2)	"kabir sharma" → "Kabir Sharma"
Swap text inside text	=SUBSTITUTE(A2, "Ltd", "Limited")	Standardising names
Split by delimiter	Data > Text to Columns > Delimited > Space	Full names into two columns

TEXTJOIN (Microsoft 365 / 2019+) joins a range with a delimiter and can skip empties — its TRUE argument — which CONCAT cannot. For building comma lists of regions that had values, TEXTJOIN is the one-line answer interviewers now expect.

TEXT

```
=TEXT(A2, "DD-MMM-YYYY")
```

TEXT converts anything to a formatted string — dates to "05-Oct-2026", numbers to "1,23,456" (with a format code), 0.156 to "15.6%". It is also the fix when a date must live inside a sentence: ="Due on " and TEXT(A2,"DD MMM") — concatenating a raw date gives you a serial number like 46266, the mistake everyone makes once.

Dates: the serial-number logic

Excel stores each date as days since 01-01-1900. Everything useful follows from that one fact:

- Subtract two dates and you get elapsed days: =A2-A1. Format as Number, or you will see a bizarre 1900 date.
- =TODAY() moves with the machine's clock: =A2-TODAY() is days since the order; negative means overdue.
- =EOMONTH(A2, 0) is the month-end date of A2 — the anchor for monthly MIS cutoffs. EOMONTH(A2, 1) is next month's end.
- =EDATE(A2, 3) is the same day three months out — due dates, warranty expiry.
- =DATEDIF(A2, B2, "d") counts days between; "m" for months (a hidden function: it works, it just is not in AutoComplete).

- Group-by-month in a pivot (Chapter 10) is built on these serials — which is why text-stored dates refuse to group.

The date trap in Indian files

05-10-2026 parsed as US reads as 10 May; as Indian it is 5 October. When data arrives mixed, Text to Columns on the date column, choose Date and DMY in the last step, and Excel re-parses every row consistently. When a test file's date column sits left-aligned, it is text — the grouping in your pivot will offer to group by nothing.

Error-reading: the skill that makes you look senior

Errors are not noise; each one is a specific diagnosis. The table below is worth memorising verbatim — interviewers literally ask “what does #REF! mean”:

Error	Meaning	Usual cause in a test file
#N/A	Lookup found nothing	Trailing space; text-number mismatch; value absent
#REF!	Reference destroyed	Deleted rows/columns a formula pointed at
#DIV/0!	Division by zero	Denominator empty or zero — wrap or test first
#VALUE!	Wrong type	Arithmetic on text; ranges of different sizes in SUMIFS
#NAME?	Unknown name	Typo, or a newer function on an old Excel
#NUM!	Impossible number	SQRT of a negative; bad iteration
#####	Column too narrow	Widen the column — not an error at all

IFERROR

```
=IFERROR(G2/H2, "n/a")
```

IFERROR returns its second argument whenever the first errors. Use it at the presentation layer; use the diagnosis when cleaning. There is also IFNA, which catches only #N/A — the precise tool when you want genuine errors (#REF!) to stay loud but absent lookups to be civilised. Knowing the difference is a genuine differentiator in interviews.

Paste Special: five minutes that save hours

- Copy, then Alt+E+S+V (Paste Values) — replaces formulas with their results. The standard finish before sending any file out.
- Alt+E+S+T — paste formats only.
- Paste Special > Operations > Multiply by 1 — the classic fix for numbers stored as text across a column.
- Alt+E+S+R — paste as transposed (rows become columns).
- Paste Special > Skip Blanks — pastes over a range without overwriting empty source cells with blanks.

DATA WORK

Chapter 9

Cleaning Messy Data

The professional's sequence

Cleaning is not tinkering; it is a checklist in a fixed order, and doing it in this order is what reviewers mean by “good data hygiene”. Every live test with a dirty file follows it:

- 1 Survey: Ctrl+End (jumps to the last used cell — far below real data? ghost cells), Ctrl+Down each column, COUNT vs COUNTA vs COUNTBLANK row by row. Note what is wrong before touching anything.
- 2 Standardise types: dates parsed as dates (Text to Columns, DMY), numbers as numbers (Multiply-by-1 paste or VALUE), one case convention for text (PROPER or UPPER).
- 3 Kill structural noise: TRIM the trailing spaces, Remove Duplicates, delete fully blank rows (filter by one column and delete the visible blanks).
- 4 Shape columns: Text to Columns to split names; Flash Fill for pattern fills; CONCAT or TEXTJOIN to join.
- 5 Verify: row counts before and after each step must be explainable. If 4,000 rows became 3,412, you must be able to say why in one sentence.

Remove Duplicates

Select the data, Data > Remove Duplicates, tick the columns that define uniqueness — on Sharma Sales that is Date + Salesman + Product + Qty. Excel reports how many duplicates it removed and how many unique rows remain; **write that number down**. The interview version: “Remove Duplicates works on a row snapshot; I keep the raw sheet untouched and clean a copy, because the operation cannot be undone after the file is saved and re-opened.”

Text to Columns: the two in one

Delimited mode splits “Sharma, Aarav” into two columns on the comma — or a full name on the space. Fixed-width mode slices positions (bank statement exports). The hidden second use: forcing types — choose Date: DMY in the final step and a text-date column becomes real dates without any formula.

Flash Fill

Type the result you want in the first cell — “Sharma, Aarav” from “Aarav Sharma”, or an email built as aarav.sharma@sharmatraders.in — press Ctrl+E, and Excel infers the pattern down the column. It is pattern-matching, not a formula: it does not update when the source changes. Interviewers ask exactly that: “Flash Fill or formula?” — formula if it must stay live, Flash Fill for a one-time re-shape.

TRIM and CLEAN

TRIM

```
=TRIM(A2)
```

TRIM removes leading, trailing and doubled spaces (it keeps single spaces between words). CLEAN strips non-printable characters that ride along in ERP and PDF-pasted data — the invisible cause of lookups that refuse to match. The thorough version on an imported column: =TRIM(CLEAN(A2)), pasted back as values.

Numbers stored as text

Three fixes, in increasing elegance: Paste Special multiply-by-1 across the column; the VALUE function (=VALUE(E2)) in a helper column; or Error Checking's Convert to Number on the green-triangle cells. Whichever you use, verify with a SUM afterwards — if the total moved, the column was text.

Go To Special

Ctrl+G > Special selects by type: Blanks (then type 0 and press Ctrl+Enter to fill every blank in one stroke — the standard move before a pivot, which treats blanks as their own group), Visible cells only (after filtering, so copying takes only what is shown), Row differences, Constants versus Formulas. Nobody teaches this in college courses; every analyst uses it weekly.

The Sharma Traders clean, end to end

Applying the whole chapter to the practice file — do this once, slowly, and the live-test version becomes muscle memory:

- 1 Ctrl+End on Sales. COUNTA the Date column (20), COUNTA Amount (0 — not built yet), COUNTBLANK Paid (3). Two duplicate rows and one “kabir ” exist by construction.
- 2 Standardise dates: select column A, Text to Columns > Delimited > (no delimiter needed, straight to step 3) > Date: DMY > Finish. All dates now right-align.
- 3 Fix the text cases: =PROPER(C2) in a helper column, paste values back.
- 4 TRIM the Salesman column with =TRIM(C2), values back. “kabir ” becomes “Kabir”.
- 5 Remove Duplicates on Date + Salesman + Product + Qty. Note the count removed.
- 6 Fill the three blank Paid cells: Ctrl+G > Special > Blanks, type No, Ctrl+Enter.
- 7 Verify: 18 unique rows, every column counts, Amount column still empty — it gets built in the Chapter 6 drills.

The sentence that closes a cleaning task

“Started with 20 rows including 2 duplicates; after cleaning, 18 unique rows, all dates parsed as DMY, no blanks in Paid, and I kept the raw sheet untouched as Sheets('Raw').” Numbers, method, and a backup habit — that is a senior-sounding close in under five seconds.

DATA WORK

Chapter 10

Pivot Tables, Zero to Confident

What a pivot table is

A pivot table takes 4,000 rows and answers “how much did each region sell, by month?” without a single formula. You drop fields into four boxes and Excel does the grouping. Interviewers for MIS and analyst roles check pivots before almost anything else, because a candidate slow in pivots is slow in the job. The full task, verbatim from real tests: “Insert a pivot table that shows total sales by region and month, add a slicer for product, and make it a pivot chart.”

Building one, step by step

- 1 Click any cell inside the data (or the Table — Ctrl+T first is better; see below).
- 2 Insert > PivotTable > New Worksheet. Excel proposes the range it found; check it — the number one pivot failure is a wrong range.
- 3 The PivotTable Fields pane appears. Four areas live at its bottom: Filters, Columns, Rows, Values.
- 4 Drag **Region** to Rows. Drag **Amount** to Values. Done — that is the region summary.
- 5 Now drag **Product** to Columns. You have a region x product matrix.
- 6 Click the Dates field, drag to Rows below Region, then right-click a date in the pivot > Group > Months (and Years if the data spans more than one). Monthly view without touching the source data.

The four areas, and how people think in them

Area	Job	Example on Sharma Sales
Rows	What each line of the summary is	Region, then Salesman
Columns	The second dimension across the top	Product, or Month
Values	What gets counted or summed	Sum of Amount, Count of orders
Filters / Slicers	Interactive narrowing of everything	Paid = Yes only

Why build on a Table first

If the source is a Table (Ctrl+T), new rows appended next month are inside the Table, and Refresh picks them up automatically. On a plain range, the pivot quietly keeps summarising the old 4,000 rows while the sheet has 4,300. That stale-pivot error reaching a manager is a rite of passage — avoid it on day one.

Changing the calculation

- Click the arrow on “Sum of Amount” in Values > Value Field Settings: Sum, Count, Average, Max, Min. Switch to Count to count orders, Average for average order value.
- Same menu > Show Values As: % of Grand Total, % of Column Total, Difference From — the region-share and month-over-month views managers actually ask for.
- Value Field Settings > Number Format formats just the pivot's numbers — the single most forgotten step; a pivot inheriting ugly decimals makes the whole file look amateur.

Grouping: dates, numbers and text

Right-click inside a pivot on any date row > Group offers Seconds through Years. Multi-select with Ctrl to group specific regions into zones, then use Group Selection for custom buckets — the North+East as “North Zone” move that sounds trivial and impresses reviewers. Numeric grouping bins a quantity column into 1-5, 6-10, 11-20 buckets. When grouping is greyed out, the field is not a real date or number — go back to Chapter 9 and fix the type.

Slicers and timelines

Click the pivot > PivotTable Analyze > Insert Slicer > tick Product. A Slicer is a row of clickable buttons that filters the pivot (and every pivot you connect to it via Report Connections). A Timeline is a slicer for dates — drag across October to November. “Add a slicer” appears in a majority of practical tests because it is the interactive half of dashboards. Format it once (two columns of buttons, a colour from the file's palette) and the file already looks like a professional build.

Pivot charts

With the pivot selected: PivotTable Analyze > PivotChart. The chart follows the pivot — click a slicer button and the chart redraws itself. This is the entire mechanic of Excel dashboards built by non-programmers. One discipline: a pivot chart shows summary shapes (columns, lines, stacked bars), not scatter plots.

Calculated fields, and their limits

PivotTable Analyze > Fields, Items and Sets > Calculated Field builds a new measure from existing ones: = Amount / Qty inside the pivot gives average price per row — without touching the source. Two limits interviewers probe: a calculated field operates on sums, so an average-of-ratios built this way is wrong (build a ratio column in the data instead); and GETPIVOTDATA formulas appear when you try to point a normal cell at a pivot result (they look hostile; they are stable references to pivot cells — usable, or disable by typing the cell reference by hand).

Refresh, and the staleness discipline

- Data changed? Right-click the pivot > Refresh. Pivots never update themselves.
- PivotTable Options > Data > Refresh data when opening the file — set it on any recurring report.
- After any Refresh on a filtered pivot, check the Filters area and slicers are still showing “all” where you mean all.
- Change Data Source re-points the pivot if rows fell outside the range — or make the source a Table and never do this again.

The 90-second pivot drill

From raw data to: region x month matrix, values as % of column, product slicer connected, one column chart. Do this daily for a week and no live test pivot task can surprise you. This exact drill is Day 17–20 of the plan.

The ten pivot tasks you will actually meet

Collected from analyst and MIS test formats and interviewer write-ups — if a practical test has pivots, it is almost always one of these ten. Each line gives the task as the test states it and the exact moves.

The task, as the test words it	Your moves
Total sales by region	Rows: Region. Values: Sum of Amount
Sales by region by month	Rows: Region; Columns: Date grouped by Month; Values: Sum of Amount

The task, as the test words it	Your moves
Count of orders per salesman	Rows: Salesman; Values: Count of Date (counting orders, not summing)
Average order value by region	Values: Average of Amount (Value Field Settings)
Region share of total	Values: Sum of Amount > Show Values As > % of Grand Total
Top 5 salesmen by revenue	Rows: Salesman; Values: Sum; Value Filters > Top 10, change to Top 5
Unpaid orders by region	Filters area (or slicer): Paid = No; Rows: Region; Values: Sum
Product mix per month	Rows: Month; Columns: Product; Values: Sum (a stacked composition view)
Quantity sold, month over month difference	Values: Sum of Qty > Show Values As > Difference From > Previous, Month
Average quantity per order by product	Helper column or Calculated Field discussion: Sum of Qty / Count of orders, done correctly per Chapter 10's calculated-field limits

Practice rule for the ten

Shuffle them: have a friend (or a random number picker) call out two tasks, and build both in one pivot in under three minutes. Tests combine tasks; practice must too.

PRESENTATION

Chapter 11

Charts and the One-Page Dashboard

Charts: the right type for each question

The question	Chart	Notes
Compare categories	Column (bar)	Regions, salesmen, products
Trend over time	Line	Months on the x-axis, always
Composition	Stacked column or bar	Share of each product in a month
Composition of one total	Pie (under 6 slices)	More slices = unreadable; use a bar instead
Relationship of two measures	Scatter	Qty vs Amount, spotting outliers
Single number in context	Big number + sparkline	The dashboard staple

Making a chart, and making it clean

- 1 Select the data (or the pivot) > Insert > Charts, choose the type. Never accept the default look — the next four steps are what reviewers judge.
- 2 Delete the chart border and gridlines you do not need; keep the horizontal gridlines on a column chart, nothing on a line.
- 3 Title states the takeaway, not the fields: “South leads Q4 sales at 34%” beats “Sum of Amount by Region”.
- 4 Data labels: add when there are few values, remove when bars are many. Legend: only when more than one series.
- 5 Colours: one accent colour for the story, grey for the rest. Excel defaults are loud; grey-quiet charts read as senior.

Chart-type traps

- Time axis must be sorted and evenly spaced; a missing month makes a line chart lie.
- A truncated vertical axis exaggerates change. In a test, start a column chart's axis at zero and mention you did.
- 3-D charts look impressive in 2007 and cost marks in 2026.
- Secondary axes are powerful and easy to misread — if a second axis is truly needed, label it directly on the line.

The one-page dashboard

A dashboard is one screen that answers five questions a manager asks every Monday. You will build this across Days 21–23; here is the target, on a sheet called **Dash**, built entirely with pivots and slicers:

- 1 Top band: three “big number” cells — total sales, count of orders, % paid — each a single pivot or formula cell with a label above it.
- 2 Left half: column pivot chart, sales by region, sliced by product.
- 3 Right half: line pivot chart, sales by month, with the year-to-date total in the title.

- 4 Bottom band: a small table of top 5 salesmen (pivot sorted descending, Value Filters > Top 10 changed to Top 5).
- 5 One slicer strip across the top: Product and Paid, both connected to every pivot on the sheet (right-click slicer > Report Connections).

Twelve charts is not a dashboard

A page readable in ten seconds is a dashboard; a page that requires scrolling is a report. Reviewers in take-home tasks consistently reward restraint — five elements, one story, one accent colour.

Sparklines and conditional shapes

Select cells beside a row of monthly numbers > Insert > Sparklines > Line: a trendline lives inside a single cell, perfect next to each region in a summary table. Win/Loss sparklines show positive versus negative months as up/down blocks. Sparklines print small and clean — the quiet upgrade that separates a workbook made this year from one made last decade.

PRESENTATION

Chapter 12

Conditional Formatting and Data Validation

Conditional formatting: rules that read the data

Select a range > Home > Conditional Formatting. The rules you will actually use in tests and at work, in order of frequency:

- **Highlight Cells Rules > Greater Than** — amounts above 50,000 jump out. The five-minute version of an exception report.
- **Top/Bottom 10 Items** — change 10 to 5; the top-5 orders by value.
- **Data Bars** — in-cell bars proportional to value; instant bar chart inside a column.
- **Colour Scales** — heat map across a region x month matrix; green-to-red on a single column.
- **Highlight Duplicates** — on a key column, the instant visual duplicate check before Remove Duplicates.
- **Icon Sets** — arrows or traffic lights against a target column; use sparingly, they multiply visual noise.

Writing your own rule

Conditional Formatting > New Rule > “Use a formula to determine which cells to format” — the step that separates users from builders. To highlight the entire row of any unpaid order: select the data, write `=H2="No"` (column locked with \$, row free — Chapter 4 revisited), choose the fill. The dollars must be exactly there; this is the most common reason self-written rules misfire.

Manage, audit, clean

- Conditional Formatting > Manage Rules shows every rule on the sheet — inherited files carry dozens of dead rules that slow them down; Clear Rules cleans a selection at once.
- Rules copy and multiply with copy-paste; a file that “formats randomly” has cloned rules fighting each other.
- Sort and filter are unaffected by conditional formats; formatting is display, not data — which also means filtering “by colour” exists but breaks silently when the rule changes.

Data validation: the feature that prevents errors

Select the Paid column > Data > Data Validation > Allow: List > Source: Yes,No. Now the column only accepts Yes or No from a dropdown. This is the two-click version of data quality, and interviewers ask for it by name in practical tests (“add a dropdown for region”).

- Whole Number with between 1 and 20 on Qty rejects absurd entries at the door.
- Custom validation with a formula — e.g. `=COUNTIF($B$2:$B$500, B2)=1` — refuses duplicate keys as they are typed.
- Input Message gives the user a tooltip on selection; Error Alert chooses whether wrong entries are blocked or merely warned.
- Circle Invalid Data (Data Validation dropdown) draws red circles around existing entries that violate a rule you just added — the audit move after applying validation to dirty legacy data.

One sentence for the interview

“Validation controls what can be entered; conditional formatting controls what gets attention — I put validation on input sheets and formatting on report sheets.”

AUTOMATION

Chapter 13

Power Query: The Refresh Button

The refresh-button skill

Power Query is Excel's engine for importing, cleaning and reshaping data from files, folders, databases and the web — recorded as steps that replay on refresh. Analyst hiring managers describe it as the part most self-taught candidates skip and exactly what turns a two-hour monthly cleaning job into a button press. It lives on the Data tab: Get and Transform Data. In Excel 2016 and later it is built in; in Excel 2010 and 2013 it was a free add-in named Power Query.

Your first query in ten minutes

- 1 Data > Get Data > From File > From Workbook (or From Text/CSV for the monthly export) — or, on the current file, From Table/Range.
- 2 The Power Query Editor opens: your data, plus an Applied Steps list on the right that is empty.
- 3 Fix the salesman column: right-click the column > Transform > Trim (kills trailing spaces). Right-click again > Transform > Capitalize Each Word.
- 4 Fix types: right-click the Date column > Change Type > Using Locale > Date, English (India) — this is where mixed 05-10-2026 inputs get parsed correctly.
- 5 Right-click the header of the blank-filled column > Replace Values: null with 0 (or Remove Empty Rows from the table menu).
- 6 Home > Close and Load To > Table on a new sheet named Clean.
- 7 Change the raw file. Data > Refresh All. Watch the whole clean replay on the new data — this is the moment the feature sells itself.

What the steps mean

Every action you take in the editor becomes one named step in Applied Steps — Removed Duplicates, Changed Type, Trimmed Text. Click a step to see the data at that moment; the M language behind it (Power Query's formula language) is visible in the formula bar if you turn it on. You are not required to write M; clicking records it. Knowing that steps are replayable, auditable and reorderable is the interview-level understanding.

The transforms that matter most

Transform	Where	What it solves
Remove Rows > Remove Duplicates	Home	The dedupe that also works on import
Split Column by Delimiter	Transform	Names, codes, compound fields
Unpivot Columns	Transform	Month-columns into rows — the single best fix for badly shaped reports
Merge Queries	Home	The Power Query VLOOKUP: join two tables on a key
Append Queries	Home	Stack Jan + Feb + Mar files into one table
Group By	Transform	Aggregate without a pivot — sums by key at load time

Unpivot, explained in one line

A sheet with months as columns (Jan, Feb, Mar...) is wrong for pivoting; Unpivot turns those column headers into a single Month column with one value per row — and Unpivot is a right-click on the selected columns. Mention it in an interview and you will be the only candidate who did.

The folder trick

Get Data > From File > From Folder on the directory where the monthly .csv exports land: Power Query combines every file in the folder into one table, and each new month's file is picked up on refresh. This single pattern — folder of monthly exports into one clean table — is the most demo-worthy 30 seconds an Excel candidate can show, and it is exactly what analysts describe when they say Power Query changed their month-end.

Limits to respect

- Power Query shapes data on load; it does not write formulas inside the sheet (that is the worksheet layer; Power Pivot sits between).
- Queries are stored in the workbook — they refresh, they do not live-update.
- Very large files need the 64-bit Excel to use real memory; 32-bit chokes on a few million rows.

AUTOMATION

Chapter 14

Speed: Shortcuts and the Test-Day Routine

Why speed is tested

Practical Excel rounds are timed more often than not — the file drops, the stopwatch starts, and “we will take the sheet back in 30 minutes”. A candidate who reaches for the mouse for every selection is visibly slower long before they make an error. Speed is not showing off; it is finishing the last task instead of being caught on the fourth.

The core set (memorise these 24)

Key	Action	Key	Action
Ctrl+C / V / X	Copy / paste / cut	Ctrl+Z / Ctrl+Y	Undo / redo
Ctrl+Arrow	Jump to data edge	Ctrl+Shift+Arrow	Select to edge
Ctrl+Home	Go to A1	Ctrl+End	Last used cell
Ctrl+Shift+L	Toggle filters	Alt+=	AutoSum
F2	Edit cell	F4	Toggle \$ locks
Ctrl+1	Format cells	Ctrl+T	Format as Table
Ctrl+9 / Ctrl+0	Hide rows / cols	Ctrl+Page Dn/Up	Next / previous sheet
Ctrl+Shift+1..5	Number formats	Alt+Enter	In-cell line break
Ctrl+E	Flash Fill	Ctrl+D / Ctrl+R	Fill down / right
Alt+H, O, I	Auto-fit column width	Ctrl+G	Go To (then Special)
Alt+E, S, V	Paste values	Alt+E, S, T	Paste formats
Ctrl+Shift++	Insert cells/row/col	Ctrl+-	Delete cells/row/col

Mouse-speed tricks that count

- Double-click the fill handle (the square at a selected cell's bottom-right): a formula with data beside it fills to the bottom instantly — no dragging across 4,000 rows.
- Double-click a column border in the header: auto-fit width. Select all columns first (click the triangle) to fit the whole sheet.
- Drag a selection's border with Ctrl held to copy it elsewhere; drag without Ctrl to move.
- Alt+drag a selection to a different sheet tab to copy across sheets.
- Right-drag the fill handle to get the fill menu (copy cells / fill formatting / series).

The test-day speed routine

- 1 First 60 seconds: Ctrl+End, Ctrl+Down the key columns, freeze the top row, note the row count. Never start typing before the survey.
- 2 Write the deliverable list from the task sheet in the file's unused area — a checklist inside the artifact you will submit.

- 3 Do easy tasks first (the sort, the SUM), lookups and pivots next, charts and formatting last — formatting is polish, not marks, until the numbers are right.
- 4 Keep one undo buffer: never delete raw data; work in a copy of the sheet.
- 5 Save every five minutes with a versioned name: SharmaTest_v1, v2. A crashed unsaved file ends the interview.

Speed calibration

The standard you are training for: from raw 20-row file to clean data + 3 pivots + 1 chart in under 15 minutes. At 4,000 rows the same routine under 30 minutes. These are the timings the drills in Chapter 16 use.

GETTING HIRED

Chapter 15

The Interview: 50 Questions, Answered

How to use this bank

Fifty questions, in the rough order fresher-to-senior rounds ask them. Do not read them — answer them, out loud, with a spreadsheet open. Interviewers consistently say the failure mode is candidates who know definitions but cannot produce a formula or a pivot when asked. After each answer, do the thing you just described, in the practice file, within thirty seconds. That combination is what you are actually training.

Foundations (1–10)

1. What is the difference between a workbook and a worksheet?

The workbook is the Excel file itself; worksheets are the tabs inside it. A new workbook opens with one sheet; files commonly carry separate sheets for raw data, calculations, pivots and the dashboard. Rename every tab something meaningful — a Sheet1 in a shared file is a navigation failure.

2. What are the main data types in Excel?

Numbers, text, dates and times (stored as numbers — days since 01-01-1900), Booleans (TRUE/FALSE), and errors. The practical skill is spotting type mismatch: a number stored as text is left-aligned and invisible to SUM and to lookups.

3. What is a cell reference? Relative vs absolute?

A reference points a formula at another cell. Relative (B2) shifts when copied; absolute (\$B\$2) stays fixed; mixed (\$B2 or B\$2) locks one dimension. Press F4 while editing to cycle through the four forms. The classic broken-formula question that follows: a tax-rate formula copied down without dollars returns zeros because it slides off the rate cell.

4. How do you freeze rows or columns?

View > Freeze Panes: Freeze Top Row for headers, Freeze First Column, or Freeze Panes with a cell selected to freeze everything above and left of it. Do it in the first minute of working on any long file.

5. How do you remove duplicates?

Select the data > Data > Remove Duplicates, tick only the columns that define uniqueness, and note the counts Excel reports (removed and remaining). Keep the raw sheet untouched and clean a copy, since the operation cannot be undone after saving. For a visual check first, Conditional Formatting > Highlight Duplicates.

6. Difference between .xls and .xlsx?

.xlsx is the current format (from Excel 2007), a compressed package that cannot store macros; .xls is the legacy binary format which can; .xlsm is the modern macro-enabled format and .xlsb the fast binary variant for very large workbooks.

7. What is a named range?

A name attached to a cell or range (select it, type the name in the Name Box). Formulas read =SUM(SalesAmount) instead of =SUM(G2:G21), tables can be named, and navigation jumps via the Name Box dropdown. They make inherited files self-documenting.

8. What is the difference between a range and a Table (Ctrl+T)?

A Table is a named, self-expanding range: it carries built-in filters and styling, auto-fills formulas down columns, keeps structured references correct as rows are added, and grows charts and pivots built on it. A plain range does none of this.

9. How do you protect a sheet or specific cells?

First unlock the cells users should be able to edit (Format Cells > Protection > untick Locked), then Review > Protect Sheet with a password; protection only bites after that step. For whole-file protection, Protect Workbook structure stops sheets being deleted or moved. Mention the order — protecting without unlocking the input cells first locks everything.

10. What is Merge and Center, and why avoid it in data?

It makes one big cell across several for titles. Inside data tables merged cells break sorting, filtering, pivots and lookups — use Center Across Selection (Format Cells > Alignment) for the visual effect without the damage.

Formulas (11–20)

11. Difference between COUNT, COUNTA, COUNTBLANK?

COUNT counts numeric cells only; COUNTA counts all non-empty cells including text and errors; COUNTBLANK counts empty cells. On a data column COUNT should equal the row count — if COUNTA is higher, some values are text.

12. COUNTIF vs COUNTIFS?

COUNTIF takes one condition (range, criteria); COUNTIFS takes pairs of ranges and conditions, all of which must hold together. =COUNTIFS(B2:B21,"North",H2:H21,"No") counts unpaid North orders. The SUMIFS counterpart reorders its arguments — sum range first — which is the classic slip.

13. Explain IF with a real use.

=IF(H2="Yes","Paid","Pending") returns Paid when H2 says Yes, else Pending. Combine with AND/OR for compound conditions, nest for bands, or use IFS on Microsoft 365 for a flatter structure. The banding discipline: write the bands down before the formula, so each IF asks exactly one question.

14. What does SUMIFS's argument order do to careless users?

SUMIFS is (sum range, then criteria pairs); SUMIF is (criteria range, criteria, then sum range). Copy-pasting between them without reordering is the most common formula error in tests. Microsoft's documentation calls this out explicitly.

15. How do you handle division by zero?

Wrap with IFERROR: =IFERROR(G2/H2, "n/a") — or test first: =IF(H2=0, "n/a", G2/H2). IFERROR suppresses every error, so prefer IFNA when only missing lookups should be quiet, leaving real errors visible.

16. VLOOKUP vs HLOOKUP?

Same function, different direction: VLOOKUP searches the first column vertically; HLOOKUP searches the first row horizontally. Both need the match column first, and both default to approximate matching when the last argument is omitted — always pass FALSE for an exact match.

17. VLOOKUP vs INDEX-MATCH — which do you use?

INDEX-MATCH can look left, survives inserted columns, and works in every Excel version; VLOOKUP is faster to write and fine for quick exact-match lookups when the key is the first column. On Microsoft 365 I default to XLOOKUP, which replaces both. Give all three with the version caveat — it is the most reliable strong answer in Excel interviews.

18. What is XLOOKUP and when is it unavailable?

=XLOOKUP(lookup, lookup array, return array, [if not found], [match mode], [search mode]) — cleaner than VLOOKUP, looks in any direction, has a built-in not-found argument and a reverse search for latest-record problems. It exists in Microsoft 365, Excel 2021 and 2024, and not in Excel 2016 or 2019.

19. What is flash fill vs a formula?

Ctrl+E infers a pattern from examples and fills a column — fast but static: it does not update when source data changes. Use it for one-time reshapes; use formulas (LEFT, MID, TEXTJOIN) when the column must stay live.

20. How do you extract year or month from a date?

=YEAR(A2), =MONTH(A2), =TEXT(A2,"MMM") for the month name, and grouping a pivot by month uses the same underlying serial. If the date is stored as text, none of these work — fix the type first (Text to Columns, DMY).

Data and pivots (21–30)**21. Walk me through building a pivot table.**

Click inside the data (or its Table), Insert > PivotTable > new sheet, then drag fields: categories to Rows, the measure to Values, a second dimension to Columns, filters or slicers to narrow. Then set the value calculation (Sum, Count, Average), the number format, and group dates by month where needed. Offer to do it live — interviewers take the offer.

22. How do you group dates by month in a pivot?

Right-click a date in the pivot > Group > choose Months (and Years if data spans multiple years). If Grouping is greyed out, the field contains text-dates or blanks — fix the column and refresh.

23. What are slicers, and how are they different from report filters?

Both filter a pivot, but a slicer is a visible set of buttons on the sheet, can filter several pivots at once via Report Connections, and is the interactive layer of a dashboard. A report filter is a dropdown buried in the pivot's own layout.

24. How does a pivot table decide what to include?

Its source range — set at creation and changeable via Change Data Source. A pivot never auto-expands a plain range; making the source a Table (Ctrl+T) means new rows enter on refresh. Stale pivots summarising old ranges are the classic inherited-file error.

25. What is a calculated field, and when is it wrong?

A new measure inside the pivot built from existing ones (Amount / Qty for average price). It operates on the aggregated values, so building a ratio of sums works, but an average of per-row ratios does not — build the ratio column in the data instead. Also expect GETPIVOTDATA to appear when referencing pivot cells; it is a stable reference, not an error.

26. How do you show percentages in a pivot?

Value Field Settings > Show Values As: % of Grand Total, % of Column, or Difference From previous for month-over-month. It is a display of the same underlying aggregation — no helper column needed.

27. A pivot shows months in the wrong order. Fix?

Custom sort the field: right-click > Sort > More Sort Options > Custom List, or fix the pivot's field settings; also check the source dates are real dates, since text-months sort alphabetically. Mention that custom lists (Jan, Feb...) are the general tool for non-alphabetic orders.

28. How do you clean a column of names with trailing spaces and mixed case?

=TRIM(PROPER(A2)), then paste values back. CLEAN additionally strips invisible control characters from ERP exports. Verify with LEN before and after — the length dropping proves the fix worked.

29. How do you split one column into two?

Text to Columns on the delimiter (comma, space), Flash Fill from typed examples, or LEFT/RIGHT/FIND formulas when the column must stay live. In Power Query, Split Column by Delimiter — replayable on refresh.

30. What is Power Query for?

Importing, cleaning and reshaping data as recorded, replayable steps: trim, case-fix, type-fix with locale, remove duplicates, merge and append tables, unpivot month-columns. It turns the monthly two-hour clean into Data > Refresh All. Get Data on the Data tab; built into Excel 2016 and later.

Charts, dashboards and quality (31–38)**31. When would you use a line chart vs a column chart?**

Line for continuous time series with many points — the shape of the trend is the message. Column for comparing a handful of categories. Stacked column when composition over time matters; pie only for a small number of shares of one total.

32. What makes a good dashboard?

One screen, five-ish elements, one accent colour, every chart titled with its takeaway, one or two slicers connecting everything, and a big-number band for the headline metrics. Ten seconds to read, one story, nothing decorative. Reviewers penalise chart-sprawl take-homes far more than plain ones.

33. How do you highlight duplicates or exceptions automatically?

Conditional Formatting > Highlight Cells Rules; for row-level highlighting use New Rule with a formula like =\$H2="No" applied to the data range — the mixed reference is what makes the whole row highlight. Data bars and colour scales for magnitude at a glance.

34. What is data validation?

Data > Data Validation restricts entry: List (dropdowns), whole numbers between bounds, date ranges, or a custom formula (for example, refusing duplicate keys). It controls what may be typed; Conditional Formatting controls what gets attention after.

35. How would you show top 5 of a long list?

In a pivot: Value Filters > Top 10, change 10 to 5. In a sheet: sort descending and take the first five, or Conditional Formatting > Top/Bottom Items. Mention both, and that Top N by one measure while showing another is where pivots beat manual sorts.

36. How do you make a chart update when data is added?

Build the chart on a Table (Ctrl+T) or a named range, not a fixed range — both expand, and the chart follows. Pivot charts on a Table-pivot do the same on refresh.

37. What is the difference between sorting a range and sorting a Table?

Sorting a selected range can leave adjacent columns unsorted if you do not expand the selection — corrupting the row relationships; a Table sorts entire rows by construction, and its headers stay in the formula bar. This question is really about row integrity awareness.

38. How do you audit an inherited, formula-heavy file?

Formulas tab: Trace Precedents / Dependents (arrows show what feeds what), Show Formulas (Ctrl+` to toggle the whole sheet), Evaluate Formula to step through one calculation, and Go To Special > Formulas to select every formula cell. Then check the pivot sources and the named ranges. Never trust a total you have not traced.

Scenario round (39–46)**39. You get 4,000 rows of messy sales data and 30 minutes. What do you do?**

Survey first: Ctrl+End for the real footprint, Ctrl+Down the key columns, COUNT vs COUNTA to find text-stored numbers, filters on to eyeball the categories. Then standardise types (dates DMY via Text to Columns, numbers via multiply-by-1), TRIM the keys, Remove Duplicates with a noted count, build the pivot the task asked for, and only then charts and formatting. Numbers first, polish last, raw sheet untouched.

40. Your VLOOKUP returns #N/A for values you can see exist. Why?

Almost always a key mismatch: trailing spaces (TRIM), number-stored-as-text on one side (VALUE or multiply-by-1), or genuinely different spellings. Test one failing pair with =EXACT() and =LEN() on both sides — that pair of checks is the professional answer.

41. A manager wants sales by region by month, refreshed monthly, in one click. Build?

Raw export in a folder > Power Query from Folder (combines files, trims, types with locale, removes duplicates) > load as Table > pivot on it with month grouping > pivot chart + two slicers on a dashboard sheet. Next month: drop the file in the folder, Refresh All. Describe this and you have described an automation, which is the point of the question.

42. Two columns of figures disagree by a small amount. How do you find the difference?

Add a difference column (=A2-B2), filter to non-zero, and eyeball the survivors; conditional formatting data bars on the difference column make the outliers visible instantly. For cross-sheet totals, SUM with full-column checks and Trace Precedents on the summary cells. The method — difference column plus filter — is what they want to hear.

43. You must send a report every Monday to non-Excel people. What do you send?

A PDF of a clean, one-page dashboard sheet (no live formulas exposed), plus the workbook behind it for anyone who asks. Paste-values any volatile layer, check the print setup (fit to one page wide, titles repeated), and make sure the numbers on the PDF match the pivots refreshed that morning — stale-PDF errors are how reports lose trust.

44. How would you count orders above Rs 50,000 that are still unpaid?

=COUNTIFS(G2:G21,">=50000",H2:H21,"No") — criteria in quotes, ranges the same size. Then volunteer the SUMIFS twin for their total value; interviewers ask the second part the moment you finish the first.

45. Data is spread across one sheet per month. Combine?

Append the sheets in Power Query (Home > Append Queries), or stack the ranges manually into one Table once if it is a one-time task. Then one pivot over the combined table. The wrong answer is twelve separate pivots and a calculator.

46. A formula shows #####. Panic?

No — the column is too narrow for the number; double-click the column border. It is not an error value, just display. Candidates who know the six real errors (#N/A, #REF!, #DIV/0!, #VALUE!, #NAME?, #NUM?) from ##### are separated in one question.

Judgement (47–50)

47. How do you make sure your numbers are right before sending?

Cross-check totals two ways (SUM of detail versus the pivot grand total), verify row counts after every cleaning step, check one known record end-to-end, and leave the raw sheet untouched so anything can be re-derived. Then, and only then, paste values for the output layer.

48. What Excel feature saves you the most time?

A defensible one-answer version: Power Query for recurring files; pivot tables for one-off questions; the double-click fill handle for daily speed. Pick one, explain the workflow it replaced, and give the time it saved — specificity is what the question tests.

49. Where do you still get stuck?

Honest, bounded, recoverable: macros beyond recording, or a heavy M transformation. Then show the habit: check Microsoft's documentation first, test on a copy, ask after twenty minutes rather than two hours. Nobody believes “nothing”.

50. What will you learn next?

Name the natural continuation of the role: SQL for anything data-adjacent (it compounds), Power BI if the company is Microsoft-licensed, or Python for larger data. Tie it to the job's own stack, not to fashion — the question is really about whether you have a direction.

GETTING HIRED

Chapter 16

The Live Test: Tactics and Ten Timed Drills

The anatomy of a live round

Knowing the format halves the anxiety. The practical round typically runs 20–45 minutes at a shared laptop or a remote screen-share, with the task sheet on paper or in chat. Some employers use a hosted assessment platform instead of Excel itself — same tasks, simulated spreadsheet. The task sheet almost always follows one skeleton: clean, summarise, look up, count conditionally, chart, interpret. Expect an observer who is not judging your speed as much as your **order**: do you survey before typing, do you keep the raw data intact, do you check your result.

Tactics that score

- Read the whole task sheet first (90 seconds) and write a numbered checklist in a far column of the sheet. Reviewers can see it; it reads as method.
- Survey the file before touching it: Ctrl+End, Ctrl+Down the key columns, note the row count, freeze the top row.
- Do the data-type fixes early — lookups and date grouping fail late if you skip them, and late failures cost composure.
- Number your sheets: 1_Raw, 2_Clean, 3_Pivot, 4_Dash. Deliver a file whose story is visible in the tab bar.
- Narrate lightly while working: “I am trimming the key column because the lookup mismatch is usually trailing spaces.” It converts silence into evidence of judgement.
- Never submit with #N/A in a visible output; either the match is genuinely missing (say so) or you fixed it. Either way, one sentence beats a red cell.
- Save versions: v1 after cleaning, v2 after pivots, v3 final. A crash then costs one stage, not the round.

Ten timed drills

Each drill uses the Sharma Traders file from Chapter 2 (built out to the row count given). Do each twice: once untimed for correctness, once against the clock. The stated target is the interview-grade time.

#	Drill	Target
1	Fill Amount = Qty x Unit Price; add GST column from a locked 18% cell; totals row	6 min
2	COUNT / COUNTA / COUNTBLANK audit of every column; two-sentence written finding	5 min
3	Band every order A/B/C/D by Amount with nested IF (or IFS); counts per band	6 min
4	VLOOKUP targets by region, wrap IFERROR, flag misses; then the same lookup with XLOOKUP	7 min
5	Full clean sequence of Chapter 9 on the dirty copy; report rows in and out	10 min
6	Pivot: sales by region x month, grouped by month, % of column, number format set	7 min
7	Slicer for Product connected to two pivots; one pivot chart; top-5 salesmen table	8 min
8	One-page dashboard sheet from drill 6 and 7 outputs with three big numbers	10 min
9	Conditional formatting: highlight unpaid rows via a formula rule; data bars on Amount	5 min
10	The full mock: fresh dirty file, task sheet of 8 tasks, no notes, one save at the end	30 min

Scoring the drills

Each drill has three gates: **correct** (the numbers reconcile against a manual check of three rows), **clean** (no stray formatting, named sheets, raw untouched), and **time** (at or under target). Pass = all three, twice in a row. When drill 10 passes at 30 minutes comfortably, you are ready for a real round; at 22–25 minutes you are in the top of a typical fresher pool.

If you fail a drill

Fail it once, mark the exact minute you lost, and repeat only that segment (the lookup, not the whole file). Drills improve fastest when the repetition is surgical, not heroic.

Remote and AI-monitored rounds

Remote practicals sometimes run on screen-share with an observer, sometimes on an assessment platform with a timer and no observer. In the platform variant there is often no narrator to impress — the artifact is everything, so the checklist-in-the-sheet and named-sheets habits matter double. Some employers now state that AI assistance is monitored or disallowed; in a monitored test, Copilot is disabled and web tools are off-limits — treat it like an exam, and be glad Chapter 14 made you fast without help.

GETTING HIRED

Chapter 17

AI Inside Excel: Copilot and Judgement

What Copilot in Excel actually does

Copilot in Excel is Microsoft's built-in assistant, available with a Microsoft 365 subscription on Windows, Mac, web and iPad, and it works on files saved to OneDrive or SharePoint with AutoSave on. Per Microsoft's documentation, it can edit worksheets (insert and modify values, apply conditional formatting, data validation, borders and styles), generate data and formulas and explain them, create charts, PivotTables and shapes, surface insights as charts, summaries, trends and outliers, highlight, sort and filter data, and fetch web information with citations. It operates in three modes — edit, plan and chat — and everything it changes stays editable by you.

In 2026 Microsoft added repeatable custom skills (define how Copilot completes a standard process), workbook rules and personalization, and change history with attribution — the enterprise-grade traceability finance teams ask for. For the job seeker the relevant facts are simpler: it exists, it needs a specific licence, it is best at formula generation, pivots, quick visual summaries and explaining inherited formulas, and it can be wrong — Microsoft's own FAQ says to review anything it produces.

Using it without becoming helpless

- Ask for the formula, then read it: “Add a column that calculates profit margin for each product” gives you a formula you must be able to defend in the interview that follows.
- Use it to explain, not only to build: “Explain the formula in cell D3” is the fastest way to decode an inherited file.
- Keep the foundations. The functions in Chapters 6–8 are the vocabulary you need to even judge whether a generated formula is right. AI raises the floor; the test still asks you to write =SUMIFS unaided.
- Ask for insight formats specifically: “Show total sales by region as a PivotTable” — vague questions get vague summaries.

The interview version of “do you use AI?”

“I use Copilot for first-draft formulas and quick summaries, and I read every output before it touches a real file — it accelerates the work, it does not replace the checking. For anything that repeats, I prefer Power Query, because it is deterministic.” That answer signals both currency and judgement, which is the exact combination employers probe for.

What not to do in a live test

If the test is monitored, Copilot and web AI are typically disabled or disallowed — treat them as unavailable by default. And in any take-home task, submitting a formula you cannot explain is a worse outcome than a slower manual build: follow-up questions in the next round will target exactly the cells you generated. Use AI to learn faster (explain, quiz me, generate practice files) rather than to produce, and the skill stays yours.

Ten prompts worth stealing

These are in the spirit of Microsoft's own documented prompt examples for Copilot in Excel, adapted to the Sharma file. The pattern in every good prompt: name the range or table, state the output format you want, and add the constraint.

Prompt	What you get
Which region had the highest and lowest sales?	Insight cards naming the extremes
Are there any outliers or anomalies in this data?	Outlier flags with reasoning
Add a column that calculates the profit margin for each product	A generated formula column, explained
Add a column that shows each region's percentage of total sales	Share column with formula you can audit
Create a PivotTable of total sales by region and month	The pivot, built for you
Show total sales by region as a column chart	Chart on the stated output
Explain the formula in cell D3	Plain-language walkthrough of an inherited formula
Review the customer feedback on this sheet and identify major themes	Theme and sentiment summary as a new column
Add a summary row with average revenue and ROI	Summary row with explained measures
Sort this table by Amount descending and highlight rows above 50,000	Sort + conditional formatting in one request

In each case the follow-up is the same and it is the part that matters: read the formula or pivot it produced, check it against two known rows, and be ready to say what it did. The prompts get you speed; the verification is the skill you are selling.

GETTING HIRED

Chapter 18

Proof: Portfolio, Resume and Certificates

Resume lines that survive scrutiny

“Advanced Excel” on a resume invites exactly one question: prove it. These lines do not:

- “Built a monthly sales dashboard in Excel (pivot tables, Power Query refresh, slicer-driven charts) summarising 40k+ rows across 4 regions.”
- “Automated a 3-hour monthly data-cleaning workflow into a one-click Power Query refresh covering 12 monthly CSV exports.”
- “Reduced reporting errors by reconciling pivot totals against SUM checks before distribution (raw sheet kept immutable).”
- “Wrote and audited lookups (XLOOKUP, INDEX-MATCH) with IFERROR handling for a 5-sheet MIS workbook used by 12 people.”

Every line names the tool feature, the scale, and the outcome — and every line must be true in your practice file, because the interview will open it. Keep one workbook whose tabs demonstrate each claim, ready to screen-share.

The certificate question

Microsoft's official credential family is Microsoft Office Specialist (MOS): **Excel Associate (Microsoft 365 Apps)** via exam MO-210, and the Office 2019-generation **Excel Associate (MO-200)** and **Excel Expert (MO-201)**. They are 50-minute proctored performance tests (you do tasks in a live app, not multiple choice), Microsoft suggests roughly 150 hours of instruction and hands-on experience before attempting, and they are scheduled through Certiport, with prices set by country. If you fail, you can retake after 24 hours.

Is it worth it? For campus placements and BPO/analytics-service recruiters who filter on keywords, a MOS Associate line helps your resume clear the filter — and it is the one Excel certificate Indian HR reliably recognises. For everything else, a shareable work product (the dashboard, the Power Query refresh, a short Loom walking through it) beats paper. Do the portfolio first; the certificate only if the roles you are targeting filter for it.

A portfolio in one workbook

- 1 One file, five tabs, uploaded anywhere you can share a link (Google Drive, GitHub, personal site).
- 2 Tab 1: Dash — the Chapter 11 one-page dashboard with slicers.
- 3 Tab 2: Clean — a before/after of the Chapter 9 clean, with the steps written in a side column.
- 4 Tab 3: Pivots — the region x month matrix with % of column and top-5 table.
- 5 Tab 4: Notes — one screen of plain sentences: what the numbers say, what you would automate next.
- 6 Tab 5: Raw — the untouched original, proving you know what immutable means.

The walkthrough is the interview. Narrate it in three minutes: the problem, the cleaning decision, the summary, the finding. Hiring managers consistently say a candidate who can defend a small real artifact outperforms a candidate with a certificate and no file.

Where to keep learning, free and official

- Microsoft Support's Excel help and learning pages — every function reference in this book lives there, always current.
- Microsoft Learn's free Excel training paths, including the MOS exam skills outlines if you take that route.
- Excel's own built-in: File > New shows templates (budgets, logs, dashboards) worth reverse-engineering.
- The practice discipline that beats them all: take any export from your own life — bank statement CSV, UPI history, phone's screen-time sheet — and dashboard it.

AFTER THE OFFER

Chapter 19

Your First Month on the Job

Week one: the file nobody wants to touch

Your first month in an MIS or analyst seat usually starts with an inherited workbook: a monster of tabs named `final`, `final2` and `FINAL_use_this`, built by three people who have left. Everything from Chapters 9 and 10 applies on day one, but the professional wrapping is new: change nothing until you understand it, and record what you change when you finally do.

- 1 Copy the file. Work only on the copy. The original is somebody's source of truth until your replacement is blessed.
- 2 Map it: for each tab, one line in a Notes sheet — what it holds, what feeds it, what it feeds (Trace Precedents and Dependents are the tools).
- 3 Check the pivots' sources and the named ranges first — they are where inherited files quietly rot.
- 4 Fix types and duplicates in the raw layer only when you must, and log each fix with the row count before and after.
- 5 Never restructure a working file in your first week. Ask what breaks if this tab is wrong; build trust before you build dashboards.

The recurring reports

By week two you will own a recurring report — sales daily, collections weekly, headcount monthly. The winning move is the Chapter 13 pattern: raw drop goes in a folder, Power Query cleans it, pivots and the dashboard read from the clean table, and your Monday becomes a Refresh All plus a check. The habits that keep you out of trouble: refresh before you read; reconcile the grand total against the previous week and explain any movement in one sentence in the email body; send the PDF, keep the workbook.

The people part

- When a manager asks for numbers “quickly”, send the correct small thing fast (a pivot screenshot) with the full thing following — never the wrong big thing fast.
- Write your formulas as if the next reader is new: named ranges, a Notes tab, no orphan helper columns on random sheets.
- Every number you send carries your name. The cross-check routine from Chapter 15, question 47, is the difference between a reputation for reliability and a reputation for excitement.
- When you automate something that used to take hours, tell your manager what you did and how long it took before — quietly, once. That is the entire art of being valued.

Where this goes next

Excel got you in the door. Within two years most analysts add SQL (it compounds for a whole career), one BI tool — Power BI is the common choice in Microsoft-licensed Indian enterprise — and, increasingly, Python for larger data. None of that is needed on day one, and all of it is easier after the spreadsheet logic in this book is solid. Finish the 30 days, pass drill 10 under 30 minutes, and go collect the job.

The last word

Every employer running an Excel test is not really testing functions. They are testing whether you can be handed something messy and unclear, impose order on it, and explain what you found. That is the job. This book taught Excel because that is the interview's language — but the skill you were building was composure with data. Carry that into the room.

VERIFICATION

Chapter 20

Sources and Further Reading

Every fact in this book, verified

No statistic, function syntax, feature claim or exam detail in this book is invented. The primary sources below were consulted in September 2026; Microsoft pages are living documentation and the current version always wins.

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- Copilot in Excel capabilities, edit/chat/plan modes, OneDrive and AutoSave requirements, and accuracy caveats: Microsoft Support, “Get started with Copilot in Excel” and “Frequently asked questions about Copilot in Excel”; data-insight and formula prompts: “Get data insights with Copilot in Excel” — support.microsoft.com.
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- Excel demand across Indian roles (51 of 93 tracked roles across 8 fields): Skilture, “Most In-Demand Skills in India” taxonomy, updated July 2026 — skilture.com.
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- Excel interview question patterns at Indian MNCs (IBM, TCS, KPMG, Accenture rounds): publicly posted interviewer compilations on LinkedIn, 2025–26.
- Role and salary ranges for MIS, data analyst, finance and operations roles: CareerScan skills report 2026 and Star AI Education hiring guide, February 2026 — indicative market ranges, not offers.

Digitalaikart publishes this playbook as an independent learning product. Microsoft, Excel, Microsoft 365, Copilot and Microsoft Office Specialist are trademarks of Microsoft Corporation; this book is not affiliated with or endorsed by Microsoft. Salary figures are indicative market ranges gathered from the public sources listed above; actual offers vary by city, company and candidate.

QUICK REFERENCE

Appendix A

The Function Dictionary

Sixty-two functions, grouped by job, each with its syntax and one line on when to reach for it. Syntax is as documented by Microsoft; functions marked (365) need Microsoft 365, Excel 2021 or later.

Math and aggregation

Function	Syntax	Use it when
SUM	=SUM(range)	Total a range; Alt+=
AVERAGE	=AVERAGE(range)	Arithmetic mean of numbers
MEDIAN	=MEDIAN(range)	Middle value; outlier-resistant
MAX / MIN	=MAX(range) / =MIN(range)	Extremes
ROUND	=ROUND(x, digits)	Fix display precision at the value level
ROUNDUP / ROUNDDOWN	=ROUNDUP(x, 0)	Force direction of rounding
SUBTOTAL	=SUBTOTAL(109, range)	Totals that respect filters (109; 9 ignores rows hidden by hand only)
SUMPRODUCT	=SUMPRODUCT(A, B)	Pairwise multiply-and-sum; conditional counting before COUNTIFS existed
ABS	=ABS(x)	Magnitude of a difference
MOD	=MOD(x, n)	Remainder; every-Nth-row logic

Counting and conditionals

Function	Syntax	Use it when
COUNT / COUNTA	=COUNT(range)	Numeric cells / any non-empty cells
COUNTBLANK	=COUNTBLANK(range)	Missing values per column
COUNTIF / COUNTIFS	=COUNTIF(range, crit)	Count matching one / many conditions
SUMIF / SUMIFS	=SUMIFS(sum, r1, c1, ...)	Conditional totals; note the order swap
AVERAGEIF / AVERAGEIFS	=AVERAGEIFS(avg, r1, c1)	Conditional means
IF	=IF(cond, then, else)	Branch logic
IFS (365)	=IFS(c1, v1, c2, v2, ...)	Flat multi-band logic
AND / OR / NOT	=AND(c1, c2)	Compound conditions inside IF
IFERROR / IFNA	=IFERROR(x, alt)	Civilise errors at the display layer
RANK	=RANK(x, range)	Position in a league table

Lookups

Function	Syntax	Use it when
VLOOKUP	=VLOOKUP(v, table, col, FALSE)	Exact match, key in first column, quick one-offs
HLOOKUP	=HLOOKUP(v, table, row, FALSE)	The horizontal twin; rare but asked
XLOOKUP (365)	=XLOOKUP(v, look, ret, [if])	The modern default; any direction

Function	Syntax	Use it when
INDEX	=INDEX(range, n)	The nth value of a range
MATCH	=MATCH(v, range, 0)	The position of v in a range
INDEX + MATCH	=INDEX(ret, MATCH(v, look, 0))	Left-lookups; version-proof
CHOOSE	=CHOOSE(n, a, b, c)	Pick by position; scenario switches

Text

Function	Syntax	Use it when
LEFT / RIGHT	=LEFT(t, n)	First / last n characters
MID	=MID(t, start, n)	A slice from the middle
LEN	=LEN(t)	Length; the space-detector
TRIM / CLEAN	=TRIM(t)	Kill spaces / invisible characters
UPPER / LOWER / PROPER	=PROPER(t)	Case conventions
CONCAT / TEXTJOIN (365)	=TEXTJOIN(" ", 1, r)	Join cells; TEXTJOIN skips blanks
SUBSTITUTE	=SUBSTITUTE(t, old, new)	Swap inside text
FIND / SEARCH	=FIND(" ", t)	Position of a character (SEARCH is case-insensitive)
TEXT	=TEXT(v, "DD-MMM-YYYY")	Format a value as text
VALUE	=VALUE(t)	Text-number to number
EXACT	=EXACT(a, b)	Case-sensitive comparison; the #N/A detective

Dates

Function	Syntax	Use it when
TODAY / NOW	=TODAY()	Live date / timestamp
YEAR / MONTH / DAY	=MONTH(d)	Parts of a date
EOMONTH	=EOMONTH(d, 0)	Month-end anchor; MIS cutoffs
EDATE	=EDATE(d, 3)	Same day, n months out
DATEDIF	=DATEDIF(a, b, "m")	Elapsed months between dates
NETWORKDAYS	=NETWORKDAYS(a, b)	Working days between; SLA metrics
WEEKDAY	=WEEKDAY(d)	Day-of-week logic

Dynamic arrays (365)

Function	Syntax	Use it when
FILTER	=FILTER(r, cond)	Return matching rows, live
SORT	=SORT(r, 1, -1)	Sorted spill of a range
UNIQUE	=UNIQUE(r)	Distinct list without Remove Duplicates
SEQUENCE	=SEQUENCE(10)	Generated index column
LET	=LET(x, expr, use)	Name intermediate steps in one formula

On the page marked (365)

Every (365) function returns #NAME? on Excel 2016/2019. Before any live test, one question to the coordinator — which Excel version — decides whether this whole section is available to you. Chapter 2 explains the polite way to ask.

BUILD THEM IN AN HOUR

Appendix B

Three More Practice Datasets

The Sharma Traders file carries you through every chapter, but tests hand you unfamiliar shapes. These three datasets are cheap to build (columns and rules only — fill 25 rows each by any means) and stretch different muscles. Every one of them reuses only the techniques from Chapters 4–13.

Dataset 1: HR register (people data)

One sheet, 25 employees: Emp ID (E-101 format), Full Name, Join Date (2019–2026 spread), Department (Sales, Finance, Ops, IT), City, Salary, Performance Rating (1–5, some blank), Active (Yes/No).

- Extract first and last names (LEFT/RIGHT/FIND or Text to Columns); salary into lakh display with a custom format.
- Tenure in years: =DATEDIF(JoinDate, TODAY(), "y") — then band into 0-1, 2-4, 5+ years with nested IF.
- Pivot: average salary by department; headcount by department x active status; a slicer on City.
- Challenge: highlight entire rows where Performance Rating is blank AND tenure exceeds 2 years — a formula rule with AND.

Dataset 2: Collections tracker (money over time)

One sheet, 25 invoices: Invoice No, Customer, Invoice Date, Due Date (EDATE from invoice date), Amount, Amount Received, Date Received (blank when unpaid), Status.

- Outstanding column: =Amount - Amount Received; Days Overdue: =IF(TODAY()>DueDate, TODAY()-DueDate, 0).
- Ageing buckets: Current / 1-30 / 31-60 / 60+ days overdue, nested IF on Days Overdue.
- Pivot: outstanding by ageing bucket x customer; the classic collections view interviewers describe.
- Challenge: conditional-format the whole table red when 60+ days overdue, and add a data-validation dropdown for Status.

Dataset 3: Monthly exports folder (the Power Query one)

Save the same Sales sheet three times as Jan.csv, Feb.csv, Mar.csv in one folder (change the dates), each with one new defect: a trailing-space salesman, a text-date column, a stray blank Amount.

- Power Query from Folder; combine; Trim the salesman column; Change Type with English (India) locale on dates; Replace null with 0.
- Load to a Table; one pivot: region x month, refreshed across all three files.
- Delete Feb.csv from the folder, Refresh All, and watch the totals move — the demonstration of determinism that wins interviews.

The one-hour rule

Build each dataset once, then redo it from a blank workbook within an hour for all three. The second build is where the knowledge becomes yours; the first is just following instructions.

AS THE TEST HANDS THEM TO YOU

Appendix C

The Ten Drill Task Sheets

Each drill in Chapter 16 now has its literal task sheet — the wording an interviewer uses, which is deliberately vague about method. Read the whole sheet, plan, then execute against the clock. Where a task says ‘state your finding’, write one plain sentence in the sheet.

Drill 1 — Amounts and GST (6 minutes)

- 1. Calculate the Amount for every row (Qty x Unit Price).
- 2. Add a GST column at 18% using the rate from a single input cell.
- 3. Add a grand total row for Amount and GST.
- 4. State the total including GST.

Drill 2 — Column audit (5 minutes)

- 1. For every column, report filled count, numeric count and blanks.
- 2. Identify which columns contain text-stored numbers or dates, and fix them.
- 3. State, in one sentence, what is wrong with this file.

Drill 3 — Banding (6 minutes)

- 1. Tag every order A (1 lakh+), B (50k+), C (10k+), D (below), by Amount.
- 2. Show the count and total Amount per band.
- 3. State which band holds most of the money.

Drill 4 — The lookup double (7 minutes)

- 1. Bring each region's Target into the Sales sheet (VLOOKUP, exact match).
- 2. Wrap the lookup so misses read 0 instead of #N/A.
- 3. Add an Achieved % column (sales / target) and flag misses below 70%.
- 4. Redo step 1 with XLOOKUP on a copy of the sheet.

Drill 5 — The clean (10 minutes)

- 1. Fix all date formats to DD-MM-YYYY.
- 2. Standardise salesman names; remove trailing spaces.
- 3. Remove exact duplicate orders; state rows in and out.
- 4. Fill blank Paid cells with No. Keep the raw sheet untouched.

Drill 6 — The pivot (7 minutes)

- 1. Pivot: sales by region, columns by month.
- 2. Show values as % of the column total.
- 3. Set the number format to two decimals with Indian grouping.
- 4. State the best and worst region-month combination.

Drill 7 — Slicers and charts (8 minutes)

- 1. Add a Product slicer and connect it to two pivots.
- 2. Create one column chart of sales by region.
- 3. Add a top-5 salesmen table (Value Filters).
- 4. Name every sheet meaningfully.

Drill 8 — The dashboard (10 minutes)

- 1. One page, three big numbers: total sales, order count, % paid.
- 2. Two charts from earlier drills, aligned and titled with takeaways.
- 3. Both slicers working on everything.
- 4. Fit to one page, print-ready.

Drill 9 — The rules (5 minutes)

- 1. Highlight the entire row of every unpaid order (formula rule).
- 2. Add data bars to the Amount column.
- 3. Add a Yes/No dropdown validation to Paid.
- 4. Circle any existing invalid entries.

Drill 10 — The full mock (30 minutes)

- 1. Clean the file (dates, spaces, duplicates, blanks). State rows in and out.
- 2. Calculate Amount and GST-at-18% from a locked input cell.
- 3. Lookup targets; compute achieved % and flag under-70% regions.
- 4. Pivot: region x month, % of column, grouped correctly.
- 5. One chart, one slicer, one top-5 table.
- 6. One written finding, one sentence, at the top of the file.
- 7. Save as v2. Raw data untouched throughout.

THE WORDS PEOPLE USE

Appendix D

Glossary

Term	Plain meaning
Absolute reference	A cell address locked with \$ signs; copying does not move it
Active cell	The cell currently selected; its address shows in the Name Box
Applied Steps	Power Query's recorded list of your transformations
Band / bucket	Grouping values into ranges with IF or pivot grouping
Big number	A single headline metric shown large on a dashboard
Calculated field	A new measure created inside a pivot from existing ones
Cell reference	A formula's pointer to another cell, e.g. B2
Conditional formatting	Formatting that appears automatically when data meets a rule
Criteria	The condition half of a SUMIF-family function
Dashboard	One screen answering the recurring business questions
Data validation	Restricting what can be typed into a cell (lists, ranges, formulas)
Delimited	Text split by a character (comma, space, pipe)
Dynamic array	A formula that spills results into multiple cells (365)
Exact match	A lookup that returns only true equality (FALSE / 0)
Fill handle	The square at a cell's corner; drag or double-click to extend
Flash Fill	Pattern-based column fill (Ctrl+E); static, not live
Formula bar	Where the selected cell's formula is shown and edited
Freeze panes	Keeping headers visible while scrolling
GETPIVOTDATA	The structured reference Excel writes when pointing at pivot cells
Go To Special	Selecting cells by type: blanks, formulas, visible only
Grouping	Collapsing pivot rows into months, years or custom buckets
Header row	The label row of a table; repeat it on every printed page
Inherited file	A workbook someone else built and you must now maintain
Lookup key	The value matched between two tables (ID, code, name)
M (Power Query)	The language behind recorded query steps
Named range	A word standing in for a cell or range
Number stored as text	A numeric value Excel treats as text; sums and lookups ignore it
Pivot table	Drag-and-drop summary of a data table
Power Query	Excel's import-clean-reshape engine; refresh replays your steps
Raw sheet	The untouched source data; never edit it directly
Refresh	Re-running a pivot or query against its source
Relative reference	A cell address that shifts as a formula is copied
Slicer	Clickable filter buttons connected to one or more pivots
Spill	A dynamic array's results filling neighbouring cells
Structured reference	Table-based reference like Sales[Qty]
Structured (Table)	The Ctrl+T object: self-expanding, named, filterable

Term	Plain meaning
Take-home task	A test where you analyse data overnight and return a one-pager
Timeline	A date-specific slicer for pivots
Trace precedents	Formula-audit arrows showing what feeds a cell
Value Field Settings	The pivot dialog for Sum/Count/Average and Show Values As